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**REVIEW ARTICLE**

# Soil Cracking in Vertisols: Mechanism, Impacts and Management Strategies

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## ABSTRACT

Black cotton soil (Vertisols) exhibit pronounced surface cracking due to their high content of smectite-rich clay minerals, which experiences intense shrink-swell cycles during alternating wetting and drying conditions. These dynamic structural changes lead to the formation of deep and interconnected cracks, significantly influencing the soil structure, water infiltration, runoff generation, soil loss and overall soil stability. Consequently, the adoption of different sustainable soil and water management practices that regulate soil moisture regimes which is essential for minimizing surface cracking and enhancing the resilience of Vertisols under variable rainfall conditions. The results revealed that these management practices substantially reduced key cracking parameters, i.e., crack depth, width, length, area and volume ranging from 3.60%–40.6%, 3.80%–46.2%, 4.40%–41.4%, 5.80%–57.0% and 18.0%–74.0%, respectively. In addition to their effects on surface cracking, agronomic, biological and soil structural management practices resulted in marked reductions in runoff (3.30%–98.3%; 1.31–39.0 mm) and soil loss (2.30%–99.7%; 0.23–9.80 t $ha^{-1}$) while enhancing water-use efficiency (WUE) by up to 22.1%. Similarly, hydrological and land-configuration practices, including broad-bed and furrow systems, ridge-furrow configurations, bunding and farm ponds reduced runoff by 6.44%–75.5% (1.65–26.8 mm), soil loss by 10.9%–98.0% (0.063–9.60 t $ha^{-1}$) and improved WUE by 3.40%–41.5%. Inclusively, this review elucidates the mechanisms governing cracking behaviour in Vertisols and underscores the critical role of soil and water management practices in mitigating surface cracking, reducing runoff and soil loss and sustaining soil health. Therefore, the adoption of these practices is therefore vital for enhancing crop productivity and long-term land resilience in semiarid and arid regions.

## 1 | Introduction

Vertisols (from Latin verto, 'turn') are widely distributed soils worldwide, occupying an estimated area of approximately 304.1 million hectares (Dudal and Eswaran 1988; Ahmad 1996) (Figure 1). Vertisols are found in a wide range of climatic conditions, from humid tropical zones to arid regions, but they are most prevalent in the semi-arid tropics (Pal, Wani, and Sahrawat 2012; IUSS Working Group WRB 2015). In India, Vertisols are primarily concentrated in the Deccan Plateau region, covering approximately 79.0 m ha of land, mainly in Maharashtra, Madhya Pradesh, Chhattisgarh, Gujarat, Karnataka, Andhra Pradesh and Rajasthan (Kadu et al. 2003; Pal et al. 2009). They are characterized by high clay contents (> 30%), distinctive shrink-swell behaviour and unique subsurface structural features known as slickensides, or strong development of vertic structures (recognized by diagnostic vertic horizon) that may be accompanied by uneven surface microtopography 'gilgai' or

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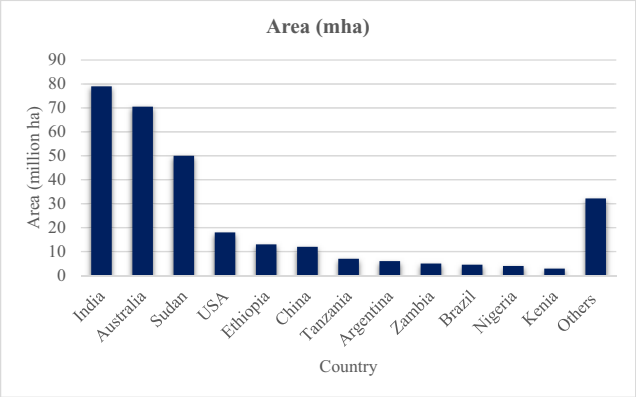



**FIGURE 1** | Global distribution of Vertisols.

wedge-shaped aggregates (Kovda et al. 2003; Pal et al. 2009). These soils, commonly known as 'black cotton soils' or 'dark cracking clays', are characterized by a high content of expansive clay minerals, mainly smectite (montmorillonite, beidellite and nontronite), which leads to significant shrinkage-swelling and changes in soil physical properties in response to moisture fluctuations (Bhattacharyya et al. 1997; Das Gupta et al. 2006; Peng and Horn 2007). This dynamic nature of Vertisols significantly affects soil structure, porosity, water movement and crop growth (Lin et al. 1997). Vertisols are widely recognized for their high productive potential; however, owing to certain physical limitations, they require specific management practices to fully realize their actual potential in the form of crop productivity (Murthy 1988; Ikitoo et al. 2011; Pal, Bhattacharyya, and Wani 2012; DeCarlo and Caylor 2019). The moisture dynamics of Vertisols pose significant challenges for farmers and researchers, as these soils are difficult to manage under both conditions, i.e., excessive wetness and prolonged dryness (Pal et al. 2014). The cracking mechanism of Vertisols is driven primarily by fluctuations in the soil moisture content followed by combined interaction of physical, hydraulic and structural processes associated with smectite-rich clay mineralogy and fluctuating soil moisture regimes. Repeated wetting-drying and freeze–thaw cycles generate shrink-swell stresses, alter aggregate stability and pore continuity, and promote the initiation, propagation and persistence of cracks, thereby strongly influencing soil hydraulic behaviour, infiltration and structural resilience (Oztas and Fayetorbay 2003). Under prolonged dry spells, intense evaporation and soil water loss cause low root penetrability and intensive shrinkage, leading to the development of wide and deep cracks (Bravo-Garza et al. 2009; Çakir and Cangir 2021). These cracks progressively widen with increasing temperature and evaporative demand and, in extreme cases, change into pothole-like depressions. The formation of potholes not only damages the soil structure but also disrupts root continuity, exposes subsoil layers, damages structures and accelerates the heavy loss of soil water through percolation or water erosion (Somasundaram, Lal, et al. 2018). Conversely, during periods of high rainfall or irrigation, these soils swell due to the hydration of clay minerals, which leads to poor drainage, reduced aeration and limited trafficability. Several factors influence drying and cracking mechanisms in Vertisols. Anthropogenic factors such as intensive tillage, high puddling, improper land configurations and monocropping systems aggravate soil cracking and accelerate the decline in productivity (Somasundaram, Lal, et al. 2018). Although Vertisols are generally neutral to alkaline because of their high base saturation and dominance of smectitic clay minerals, their chemical properties vary considerably depending on climate, parent material, drainage conditions and land-use history. In arid and semi-arid regions, salinity and sodicity frequently occur in Vertisols and may adversely affect soil structure, hydraulic conductivity, swelling-shrinkage behaviour and crop productivity (Balpande et al. 1996; Pal, Wani, and Sahrawat 2012). In contrast, acidic Vertisols have also been reported in high-rainfall and strongly leached environments such as the Ethiopian highlands and parts of Australia, where prolonged weathering, nutrient leaching, and intensive cultivation can promote soil acidification despite the inherently high clay content and base status of these soils (Ahmad 1996).

A wide array of conservation practices, including both in situ and ex situ approaches, are effective in minimizing surface cracking and maintaining a favourable soil physical environment in Vertisols (Singh 2015; Somasundaram et al. 2016; Anantha et al. 2021). These practices can broadly be categorized into agronomic, biological and soil structural management practices and hydrological and land-configuration practices based on their primary mode of action in controlling shrink–swell behaviour through improving soil moisture dynamics. Former practices such as mulching, green manuring, conservation agriculture, crop diversification or legume-based intercropping, integrated nutrient management, and the application of soil amendments (e.g., gypsum), directly influence crack formation and propagation in Vertisols. These practices not only reduce evaporation losses but also enhance soil organic matter, improve aggregation and strengthen soil structural stability, thereby reducing shrinkage stresses during drying (Bandyopadhyay et al. 2003; Hati et al. 2006; Hulugalle, Weaver, and Finlay 2010; Lopez-Bellido et al. 2016; Rahman et al. 2017). Similarly, hydrological and land-configuration practices primarily function by improving rainwater conservation and regulating soil moisture distribution within the soil profile. These are broad-bed and furrow systems, ridges and furrows, compartmental bunding, graded bunding, tied ridges, trenches, farm ponds and check dams play an important role in enhancing in situ rainwater harvesting, minimizing runoff and improving soil moisture storage (Pathak et al. 2011; Vekariya et al. 2015; Rao et al. 2022). By moderating soil moisture fluctuations, these measures indirectly reduce the intensity and persistence of surface cracking. Furthermore, the integration of agroforestry and agri-horticultural systems provides additional ecological and hydrological benefits through improved microclimatic regulation, enhanced biodiversity, increased moisture content, and reduced runoff and soil erosion (Fahad et al. 2022). Supporting this, a meta-analysis by Muchane et al. (2020) reported that agroforestry systems exhibited 75.0% higher infiltration rates and 57.0% lower runoff rates than crop monocultures, highlighting their potential role in sustainable moisture conservation and soil structural improvement.

Despite the importance of these management interventions, information on the cracking and swelling behaviour of Vertisols in agricultural systems and the effectiveness of different management practices in reducing surface cracking remains fragmented and limited. Therefore, the main objectives of this study are: (i) to understand the cracking and swelling behaviour of Vertisols; (ii) to examine the influence of soil and water



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management practices on crack development and shrink-swell dynamics; and (iii) to evaluate the effectiveness of different management interventions in reducing runoff and soil loss in Vertisols. Accordingly, this systematic review synthesizes research on the cracking and swelling dynamics of Vertisols, with particular emphasis on soil moisture management strategies aimed at minimizing surface cracking and reducing runoff and soil erosion for sustainable management.

## 2 | Materials and Methods

The review was carried out via a structured protocol aligned with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (Page et al. 2021) to ensure transparency, reproducibility and methodological rigour using major scientific databases, including Google Scholar, PubMed, ScienceDirect (Figure 2). Relevant studies were identified using a combination of keywords and Boolean search strings such as: ('Vertisol*' OR 'black cotton soil*' OR 'shrink-swell soil*' OR 'expansive clay soil*' OR 'expansive soil*' OR 'swelling clay soil*') AND ('soil cracking' OR 'surface crack*' OR 'shrinkage crack*' OR 'desiccation crack*' OR 'crack morphology' OR 'crack dynamics' OR 'crack geometry' OR 'crack parameter*' OR 'crack volume' OR 'soil shrinkage' OR 'soil swelling' OR 'shrink-swell behaviour' OR 'swelling behaviour' OR 'smectite') AND ('soil and water conservation' OR 'conservation agriculture' OR 'mulching' OR 'residue retention' OR 'green manure' OR 'organic amendment*' OR 'gypsum' OR 'broad bed and furrow' OR 'BBF' OR 'ridge furrow' OR 'bunding' OR 'compartmental bunding' OR 'raised-sunken bed' OR 'farm pond' OR 'subsurface dam' OR 'underground dam' OR 'sand dam' OR 'water harvesting structure*' OR 'check dam') AND ('runoff' OR 'soil loss' OR 'erosion' OR 'infiltration' OR 'preferential flow' OR 'groundwater recharge' OR 'soil moisture conservation' OR 'water use efficiency' OR 'hydrological process*' OR 'soil hydraulic properties').

All retrieved articles were systematically screened, and studies were retained on the basis of the following inclusion criteria: (i) The study was conducted on Vertisols or soils explicitly classified as Vertisols (e.g., black cotton soils, shrink-swell clays with smectite dominance), (ii) the study quantitatively evaluated soil cracking parameters, including but not limited to crack depth, width, length, area, volume, crack density or crack intensity, (iii) the study examined sustainable soil and water management practices, including agronomic, biological and soil structural management practices such as mulching or residue management, conservation tillage with organic matter additions, nutrient management, soil amendments and/or hydrological and land-configuration practices such as broad-bed and furrow systems, ridge-furrow systems, bunding,

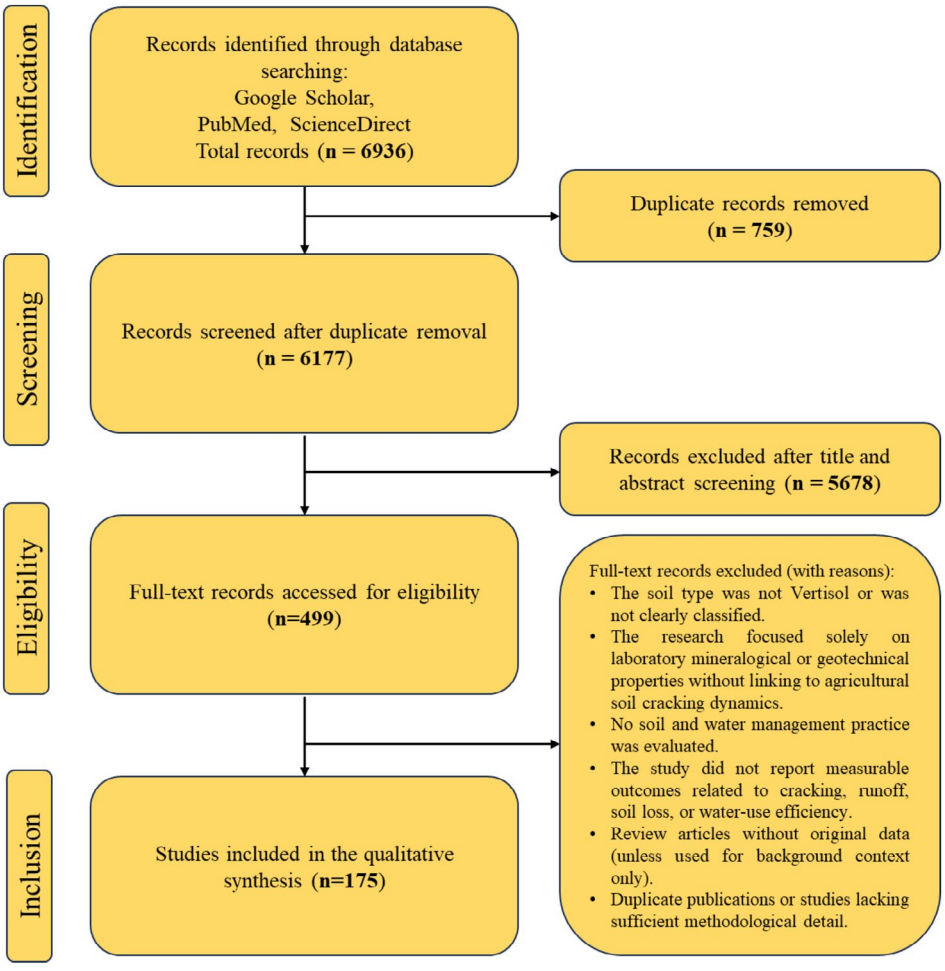



**FIGURE 2** | Overview of the literature search and study selection process.



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contouring, ponds, subsurface dams or water harvesting structures; and (iv) the study reported at least one of the following outcomes: changes in cracking behaviour, runoff reduction, soil loss or erosion control and water-use efficiency (WUE). After rigorous screening, manuscript is supported by a total of 230 references carefully identified through a systematic literature search. Out of these, 175 references are original research articles forming the primary evidence base of this manuscript. These peer-reviewed experimental and field-based studies directly contribute to the core findings, data synthesis and conclusions presented herein.

## 2.1 | Cracking Mechanism of Vertisols

Cracking in Vertisols is a defining pedogenic and hydro-physical feature that arises from the interaction of physical, hydraulic and structural processes, all of which are closely linked to the dominance of expansive clay minerals and strong seasonal moisture fluctuations (Figure 3). These processes operate simultaneously, but their relative importance varies with climate, soil management and land use.

i. Physical processes: Cracking in Vertisols is primarily governed by the high content of expansive smectitic clays (generally >30%), which undergo pronounced swelling upon wetting and shrinkage during drying, a process commonly referred to as argilli-pedoturbation (El Abedine and Robinson 1971; Dasog et al. 1988). The large specific surface area and expandable interlayer structure of smectite minerals produce substantial moisture-dependent volume changes within the soil matrix. As soil water content declines, the soil follows characteristic shrinkage curves in which progressive moisture loss results in volumetric shrinkage and increasing soil strength. When water content decreases below the plastic limit, tensile stresses develop rapidly because shrinkage occurs non-uniformly throughout the soil profile. Once these stresses exceed the tensile strength of the soil, the matrix fractures and interconnected cracks are formed (McGarry and Malafant 1987; McGarry and Daniells 1987; Morris et al. 1992; Bordoloi et al. 2020; Liu et al. 2024). According to Wei et al. (2020) cracking in clayey soils initiates when tensile stresses generated by capillary suction during water evaporation exceed the tensile strength of the soil, triggering crack formation on the surface in sequential and hierarchical patterns.

The seasonal evolution of cracking in Vertisols is therefore strongly regulated by shrinkage characteristics, soil

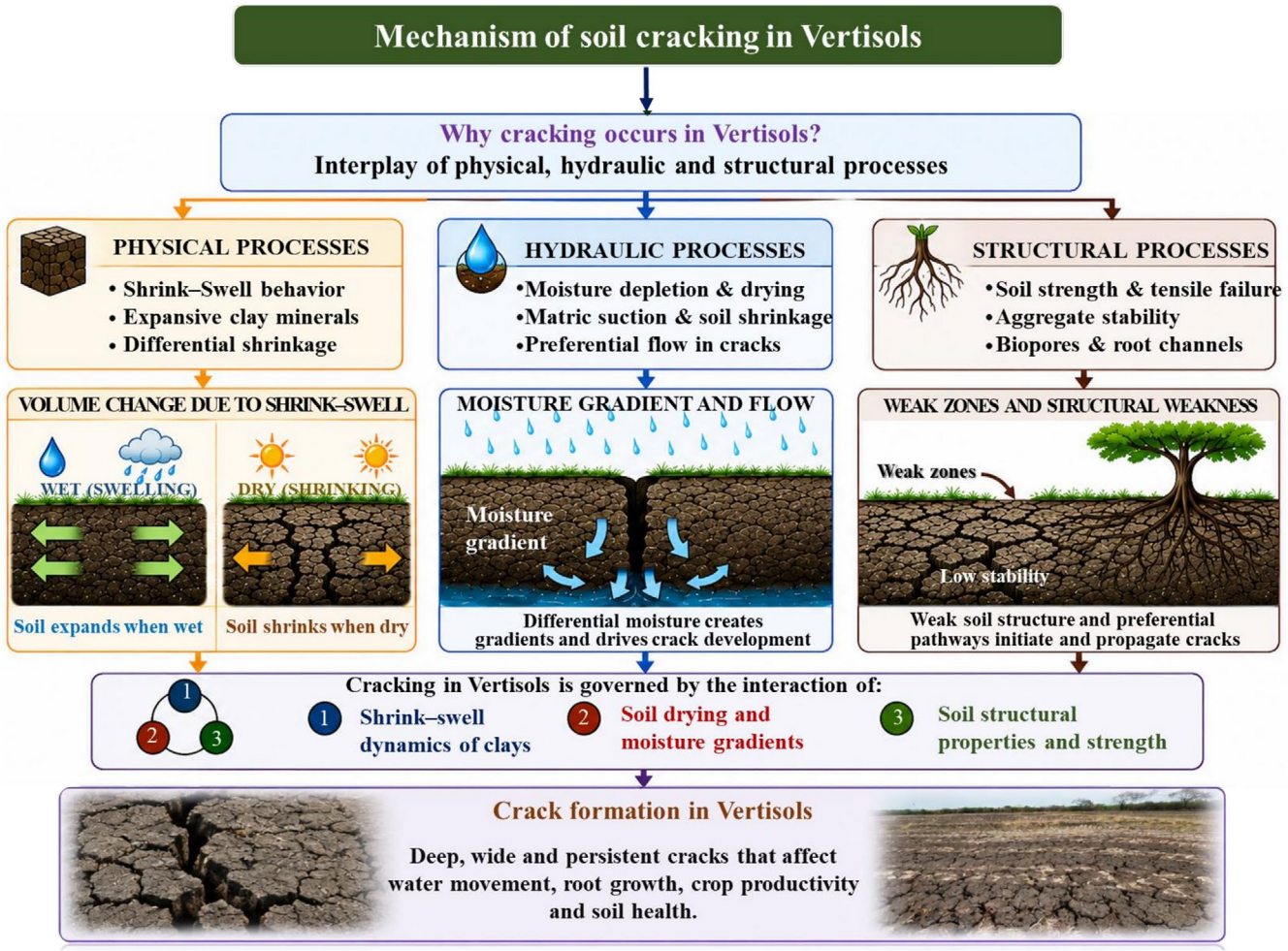



**FIGURE 3** | Schematic representation of the key processes responsible for crack initiation and propagation in Vertisols. The visual presentation of the figure was enhanced using Canva AI (Canva), an AI-assisted design tool.



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mechanical behaviour, hydraulic processes and moisture status relative to the plastic limit. During dry periods, open desiccation cracks facilitate rapid preferential flow into deeper soil layers, bypassing portions of the surface matrix and enhancing temporary subsoil wetting. As wetting progresses, swelling of smectitic clays causes crack closure, reduced macropore continuity and a transition from preferential crack flow to matrix-dominated infiltration. Under these conditions, surface sealing, ponding and runoff generation may increase substantially (Kishné et al. 2010). Repeated wetting–drying cycles further modify aggregate arrangement, pore continuity and bulk density, progressively influencing crack persistence and soil structural behaviour over time.

Soil mechanical conditions near the plastic limit are particularly important for Vertisol management. Australian studies demonstrated that tillage and machinery traffic at water contents wetter than the plastic limit, including near field capacity, can induce severe smearing, compaction, pore collapse and structural deformation because the soil remains highly plastic and mechanically unstable (McGarry and Daniells 1987). In contrast, tillage below the plastic limit allows the soil to behave in a more friable manner, reducing structural damage. Under prolonged drying, Vertisols commonly develop dense and hard-setting aggregates with reduced intra-aggregate porosity, while bulk density may increase substantially as shrinkage progresses (Srivastava et al. 1989). These interacting shrink–swell, hydraulic and mechanical processes collectively regulate infiltration, runoff, evapotranspiration, root penetration, erosion and solute redistribution in Vertisols, thereby exerting strong control over soil physical functioning and crop management under both rainfed and irrigated conditions.

In addition to wetting-drying cycles, freeze–thaw processes can substantially influence aggregate stability, crack development and structural dynamics in shrink-swell soils, i.e., Vertisols of temperate and cold regions. During freezing, the expansion of water within soil pores generates internal mechanical stresses that disrupt aggregate arrangement and alter pore continuity. Subsequent thawing weakens interaggregate cohesion and may increase structural instability. Repeated freeze–thaw cycles often accelerate aggregate breakdown, modify crack morphology and enhance the development of preferential flow pathways, thereby influencing infiltration and soil hydraulic behaviour (Staricka and Benoit 1995; Oztas and Fayetorbay 2003). Several studies have reported that cyclic freezing and thawing can deteriorate soil aggregation and reduce structural stability in clay-rich soils (Hanay et al. 2003; Sahin et al. 2008; Hewage et al. 2023). However, some study documented that freeze–thaw cycling process strengthens particle bonding, which causes an overall increase in aggregate stability of soil, showing a process from destruction to reconstruction (Kværnø and Øygarden 2006; Zhang et al. 2016; Leuther and Schlüter 2021). In Vertisols, the interaction between freeze–thaw and wetting-drying processes may further intensify shrink–swell stresses, affecting crack persistence, moisture distribution and long-term soil physical resilience under fluctuating climatic conditions (Mao et al. 2025).

Cracks can extend beyond the zone of slickensides in arid conditions, and the extent and rate of crack development are governed by multiple factors, including soil properties such as clay content, mineralogy, available water, porosity and bulk density, wetting-drying cycles, temperature fluctuations, topography, vegetation cover, cropping systems, microclimatic conditions and soil management practices (Morris et al. 1992; Azam et al. 2000; Thomas et al. 2000; Kishné et al. 2010; Dinka and Lascano 2012).

ii. Hydraulic processes: These processes strongly regulate the initiation, depth and continuity of cracks by controlling the soil water distribution and drying rates. In Vertisols, rapid evaporation from the surface during dry periods leads to steep moisture gradients between the surface and subsurface layers. This differential drying causes greater shrinkage near the surface, which initiates crack formation that then propagates downward as drying progresses (Chertkov and Ravina 1999). Once formed, cracks act as preferential flow paths during rainfall or irrigation, allowing rapid infiltration of water into deeper layers. This bypass flow can delay uniform rewetting of the soil matrix, leading to uneven swelling and further crack persistence. Moreover, repeated cycles of rapid drying and preferential wetting reinforce crack networks, increasing crack depth and volume over time.

iii. Structural processes: The development and evolution of cracking in Vertisols are closely associated with soil aggregation, structural fabric and pore continuity. Vertisols commonly possess strong blocky to wedge-shaped aggregates separated by slickensides formed through repeated shrink-swell cycles and shear deformation. These structural planes act as preferential zones for crack initiation and reopening during drying. Cracking in Vertisols results from the combined interaction of physical shrinkage, hydraulic processes and soil structural characteristics under fluctuating moisture regimes (McGarry and Daniells 1987; Kishné et al. 2010). According to Tang et al. (2021), desiccation cracking in soils is a coupled process involving water evaporation, suction increase and volumetric shrinkage, with crack initiation, propagation and coalescence. During drying, soil volume reduction generates tensile stresses, while hydraulic gradients influence stress distribution and crack propagation. Simultaneously, soil structure controls the location, orientation and connectivity of cracks. Under wet conditions, swelling may induce shear movement and slickenside formation without visible cracking (Kabala et al. 2015). In contrast, progressive drying reduces intra-aggregate porosity, promotes surface subsidence, and generates interconnected shrinkage cracks that propagate downward through the soil profile (Greco 2002; Stewart et al. 2016) (Figure 4). Therefore, understanding cracking behaviour in Vertisols requires an integrated assessment of clay mineralogy, moisture dynamics, soil structural conditions and management history, which collectively regulate water movement, root growth and overall soil physical functioning.



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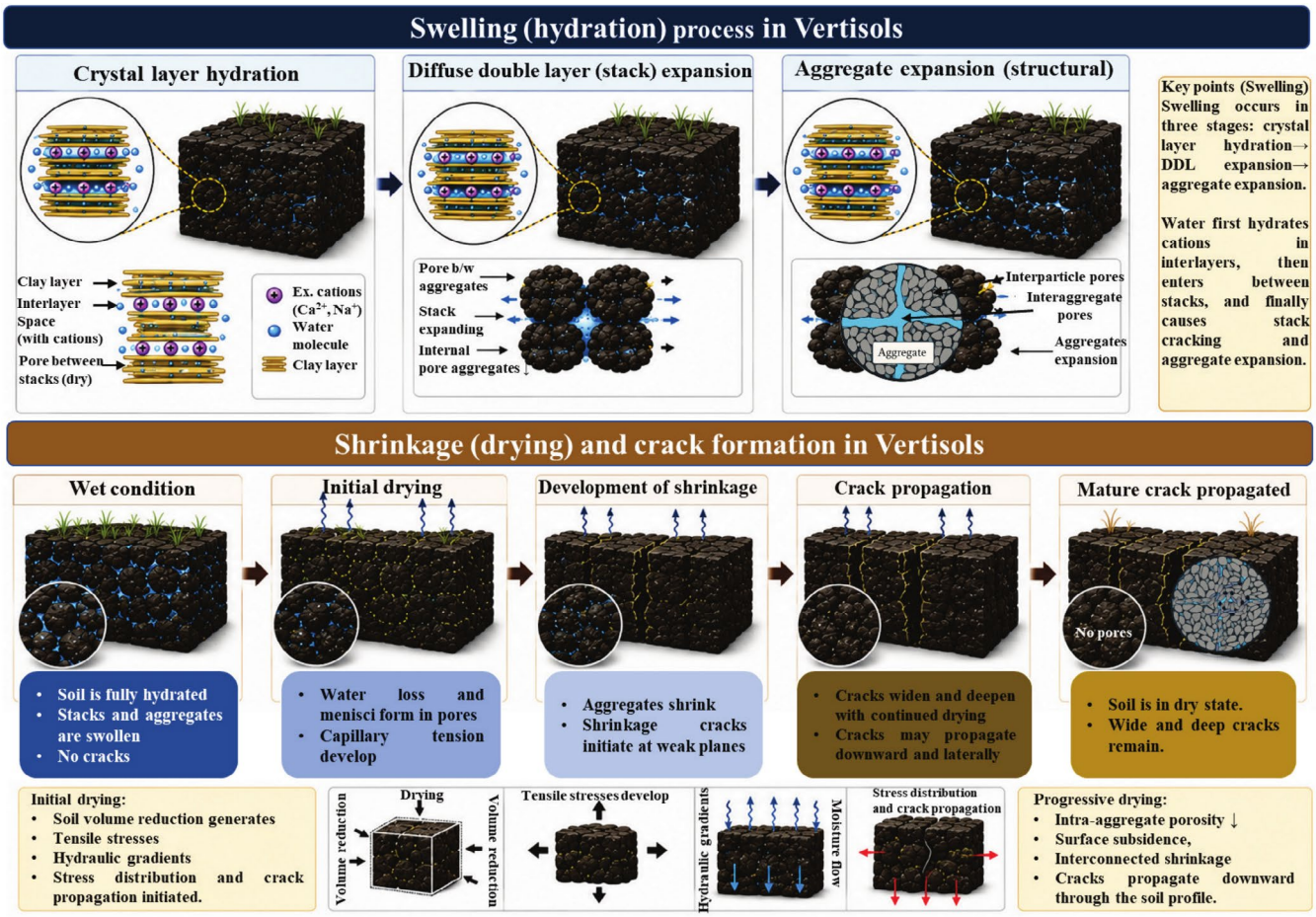



**FIGURE 4** | Schematic illustration of the swelling and cracking behaviour of *Vertisol* under different moisture regimes. The visual presentation of selected elements of the figure was partially enhanced using Canva AI (Canva), an AI-assisted design tool.

## 3 | Factors Affecting the Cracking Processes of Vertisols

The shrinking and swelling behaviour of Vertisol is influenced by several interrelated factors:

i. Clay mineralogy: Smectite clays, especially montmorillonite, have high swelling and shrinkage properties (Borchardt 1989; Shirsath et al. 2000), and secondary minerals such as kaolinite, micas, chlorites, palygorskite and vermiculites are also associated with these clays (Heidari et al. 2008). The vertic properties of Vertisol are a function of the smectite content (Bhattacharyya et al. 1997). However, palygorskite is another clay mineral that breaks apart, and its fibres do not join together to form aggregates in soil or in suspension, even when calcium ions ($Ca^{2+}$) are present (Neaman and Singer 2004). Therefore, palygorskite particles tend to move deeper into the soil profile more easily than smectite does and can eventually clog soil pores. As a result, Vertisols have high exchangeable magnesium percentages, which lead to strong dispersion of clay particles and disrupt the 3D structure of the soil matrix (Vaidya and Pal 2002; Pal, Wani, and Sahrawat 2012). This behaviour results in lower drainage when these soils are irrigated, making crop production difficult. Therefore, the amount and type of clay minerals strongly affect the intensity and extent of both horizontal and vertical cracking in Vertisols.

ii. Soil moisture dynamics: The rate of moisture loss through evaporation and transpiration governs the rate of drying. Elias et al. (2001) reported that Vertisols with higher clay content (57%–65%) developed deeper, wider and larger crack volumes than soils with lower clay content (40%–45%), highlighting the critical role of clay mineral content and seasonal wetting-drying cycles in determining crack severity. Rapid drying accelerates shrinkage, resulting in deeper and wider cracks. The crack intensity factor (CIF), a measure of crack volume relative to soil volume, increases with decreasing moisture. Lepore et al. (2009) and Kishné et al. (2010) reported that the cracking process is a function of antecedent soil moisture.

iii. Cation saturation and electrochemical stability: The shrink–swell behaviour and cracking processes of Vertisols are strongly influenced by cation composition, electrolyte concentration and electrochemical stability of the soil system. In arid and semi-arid environments, precipitation of calcium and magnesium carbonates may progressively increase sodicity, thereby impairing hydraulic conductivity, enhancing clay swelling and dispersion, and intensifying cracking processes (Pal, Wani, and Sahrawat 2012). Application of gypsum improves



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soil structural stability by increasing $Ca^{2+}$ concentration in the soil solution, promoting cation exchange, reducing clay dispersion, and improving infiltration and hydraulic conductivity. However, recent studies suggest that soil structural degradation cannot be explained solely by exchangeable sodium percentage (ESP) or sodium adsorption ratio (SAR). Sumner (1993) demonstrated that clay swelling and dispersion are regulated by the interaction between exchangeable cations and electrolyte concentration, particularly the balance between total electrolyte concentration (TEC) and critical flocculation concentration (CFC). Under low electrolyte conditions, even relatively low sodium levels may destabilize clay particles and induce dispersion, resulting in crusting, runoff, erosion and structural degradation.

To better describe these electrochemical interactions, Rengasamy and Marchuk (2011) proposed the cation ratio of soil structural stability (CROSS), which incorporates the differential dispersive effects of Na and K and the flocculating effects of Ca and Mg. Marchuk and Rengasamy (2012) further demonstrated that CROSS was more strongly related to clay dispersion and hydraulic behaviour than SAR across soils differing in texture, mineralogy and organic matter content. More recently, Rengasamy and Tavakkoli (2026) emphasized that the dispersive behaviour and structural stability of cracking soils are governed by complex interactions among electrolyte concentration, cation balance, clay mineralogy and moisture conditions. Collectively, these studies indicate that cracking behaviour in Vertisols is controlled not only by sodium concentration but also by broader electrochemical processes regulating clay flocculation, dispersion and soil structural stability.

iv. Bulk density and porosity: Bulk density, aggregation and pore distribution strongly influence shrinkage and cracking behaviour in Vertisols. Poor aggregation and high bulk density restrict water movement and root penetration, while well-aggregated structures improve pore continuity and reduce crack development. Shrink-swell behaviour and crack dynamics are closely related to soil microstructure and intra- and interparticle porosity (Coulombe et al. 1996). Although higher bulk density generally reduces shrinkage capacity because of lower pore volume, dense soils may still develop substantial tensile stresses during drying that promote crack initiation and propagation (Berndt and Coughlan 1977; McGarry and Malafant 1987).

v. Soil organic matter influences the cracking behaviour of Vertisols mainly through its effects on aggregation, porosity, moisture retention and soil structural resilience. Organic matter may improve aggregate cohesion and moderate moisture fluctuations, thereby reducing shrinkage stresses and crack persistence during drying. Repeated shrink–swell cycles may also redistribute organic matter within the soil profile through crack-mediated pedoturbation processes, contributing to gradual profile homogenization (Pierre et al. 2019). However, in highly smectitic Vertisols, crack development is governed primarily by clay mineralogy and soil moisture dynamics rather than by SOC alone. Consequently, the influence of organic matter on soil physical properties is often limited and strongly dependent on soil chemical conditions and long-term management history. Schweizer et al. (2022) reported that higher application of organic amendments in a semiarid Vertisol did not significantly increase SOC or improve aggregate stability compared with integrated fertilization, indicating that soil chemical properties may exert stronger control over structural behaviour than SOC alone. Similarly, an Australian long-term study demonstrated that stubble retention and stubble burning did not significantly alter SOC contents in Vertisols even after 50 years, suggesting that biomass addition does not necessarily result in greater SOC accumulation in highly clayey shrink–swell soils (Hati et al. 2021).

vi. Land use and management practices: Tillage, irrigation and cropping practices alter soil moisture regimes and structure, thus influencing the crack development process (Somasundaram, Singh, et al. 2018).

vii. Crop diversification: Root distribution differs among vegetation types, impacts evapotranspiration and controls the rate of water uptake from the soil, which affects soil shrinkage. On the other hand, roots can also modify crack size and distribution by binding soil particles together, increasing aggregation and reducing cracking (Mitchell and van Genuchten 1992).

## 4 | Shrinkage Dynamics, Structural Deformation and Land Degradation Processes in Vertisols

### 4.1 | Shrinkage Curves and Structural Deformation

The shrinkage and cracking behaviour of Vertisols is primarily governed by the dominance of expansive smectitic clay minerals, which undergo repeated swelling upon wetting and contraction during drying. As soil moisture decreases, volumetric contraction generates tensile stresses within the soil matrix, and cracks develop when these stresses exceed the tensile strength of the soil (McGarry and Malafant 1987; McGarry and Daniells 1987). Shrinkage behaviour in these soils is commonly described using the soil shrinkage characteristic curve (SSCC), which relates changes in soil water content to volumetric deformation and pore-space redistribution. Four major shrinkage phases are generally recognized: structural shrinkage, proportional (basic) shrinkage, residual shrinkage and zero shrinkage (Bronswijk 1991). During structural shrinkage, large interaggregate and biological pores empty with minimal bulk volume change, whereas proportional shrinkage involves nearly equivalent reductions in soil water volume and soil volume. As drying progresses into the residual shrinkage phase, air enters intra-aggregate pores and shrinkage becomes progressively less proportional. Under zero shrinkage conditions, soil particles attain their densest arrangement and further water loss produces negligible bulk volume reduction, although microscopic crack formation may continue (Cornelis et al. 2006).



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These shrinkage processes strongly influence crack geometry, preferential flow, subsidence and water redistribution in Vertisols. Stewart et al. (2016) reported that under wet conditions most pore space remains within aggregate domains as interparticle and interaggregate pores, whereas progressive drying transfers this porosity into extra-aggregate domains expressed as subsidence zones and interconnected shrinkage cracks. Thus, crack formation represents a redistribution of pore space from intra-aggregate pores towards macroscopic crack networks. Crack initiation and propagation are further controlled by antecedent moisture conditions, pore geometry, soil fabric and structural continuity. Kishné et al. (2010) demonstrated that crack development depends not only on prevailing soil moisture conditions but also on previous wetting history, indicating strong hysteresis effects associated with repeated wetting–drying cycles and long-term structural reorganization in Vertisols. Consequently, soils subjected to wetter antecedent conditions often develop greater crack area density and higher shrinkage intensity than soils with similar moisture contents but lower antecedent wetness. Structural deformation processes such as puddling, smearing, compaction, subsidence and pore collapse further modify shrinkage behaviour and crack morphology. Puddling and intensive tillage disrupt stable aggregates and disperse clay particles, while smearing caused by tillage or machinery traffic seals pore channels and reduces macropore continuity (Bandyopadhyay et al. 2003; Mohanty et al. 2004; Pendke et al. 2025). These processes create compacted layers with restricted drainage and uneven moisture redistribution, thereby increasing shrinkage stresses during drying. As a result, compacted and smeared soils often exhibit greater subsidence, pore collapse and deeper or wider crack systems compared with well-structured soils.

### 4.2 | Hydrological Consequences of Cracking

Cracking substantially alters the hydrological functioning of Vertisols by influencing preferential flow, infiltration behaviour, runoff generation, sediment transport and soil moisture redistribution. During dry periods, open cracks act as preferential pathways that rapidly transmit rainfall into deeper soil horizons, bypassing portions of the soil matrix and altering normal infiltration processes. This rapid non-matrix flow often results in uneven soil moisture distribution and enhanced downward movement of nutrients, fertilizers, contaminants and agrochemicals below the effective root zone, thereby reducing nutrient-use efficiency and increasing environmental risks (Paltineanu 2001; Kurtzman et al. 2016). Hydrological responses in Vertisols vary considerably depending on the seasonal status of crack opening and closure. Inoubli et al. (2016) reported that runoff was substantially greater under non-cracked conditions, whereas rainfall events occurring during cracked conditions frequently produced little or no runoff despite high rainfall intensity. However, once cracks swell and close following rainfall, infiltration declines sharply and runoff generation increases substantially. Crack-induced weakening of aggregates also increases soil erodibility and sediment transport even under comparatively lower runoff conditions. DeCarlo and Caylor (2019) further observed that larger crack apertures and deeper crack systems enhance water and vapour fluxes within the soil profile, thereby intensifying moisture loss and altering hydrological processes in Vertisols.

### 4.3 | Structural Degradation and Self-Repair Processes

Excessive shrinkage and cracking may adversely affect the structural integrity and long-term stability of Vertisols by weakening aggregate stability, disrupting pore continuity and creating mechanically unstable surfaces. Deep and interconnected cracks fragment the soil into large clods, impair seedbed preparation, restrict root-soil contact and reduce efficient uptake of water and nutrients. Repeated crack opening and closure may further destabilize aggregates and create persistent 'soil memory' effects in which previously formed cracks become preferential zones for future crack propagation (DeCarlo and Caylor 2019). Open cracks also increase aeration and exposure to high temperatures, thereby accelerating soil organic matter decomposition and reducing biological binding agents responsible for aggregate stabilization. However, shrinkage and cracking in Vertisols should not be interpreted solely as degradative processes. Several Australian studies demonstrated that repeated wetting-drying cycles can contribute positively to structural regeneration and self-repair in compacted Vertisols (McHugh et al. 2009; Antille et al. 2016; Isbell 2016; Niaz et al. 2023). Pillai and McGarry (1999) reported that repeated wetting-drying cycles improved clod porosity, enhanced pore connectivity and promoted formation of smaller and better-aggregated peds. Similarly, Pillai-McGarry and Collis-George (1990) demonstrated that multiple wetting-drying cycles were necessary to initiate measurable structural recovery in Vertisols, while simulated flood irrigation promoted more rapid structural repair than rainfall-driven wetting (Sarmah et al. 1996). These findings indicate that moderate shrinkage and cracking can contribute to self-mulching behaviour, alleviation of compaction, improvement of macroporosity, and enhanced infiltration and root penetration under suitable moisture and crop management conditions.

### 4.4 | Implications for Land Degradation and Management

The combined effects of shrinkage dynamics, structural deformation and hydrological alteration strongly influence land degradation processes in Vertisols. Excessive cracking may increase runoff, erosion, preferential solute transport, nutrient depletion, structural instability and drought susceptibility, thereby reducing long-term soil productivity and resilience (Dinka et al. 2013). Balpande et al. (1996) reported that poor hydraulic conductivity associated with sodicity further aggravates degradation processes by restricting downward water movement, promoting irregular carbonate accumulation and increasing deep crack formation beyond slickenside zones. Such conditions impair permeability, restrict root penetration and increase susceptibility to hardsetting and subsoil degradation. Moreover, structural degradation associated with cracking reduces the capacity of Vertisols to store and supply water efficiently, thereby increasing vulnerability to drought and nutrient loss. At the



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same time, appropriate management practices can substantially moderate shrinkage intensity and improve structural resilience in Vertisols. Practices such as mulching, residue retention, conservation agriculture, broad-bed and furrow systems, compartmental bunding and integrated nutrient management improve in situ moisture conservation, reduce evaporation losses, moderate shrinkage stresses and maintain pore continuity (Bandyopadhyay et al. 2003; Pathak et al. 2011; Patil et al. 2016). Therefore, sustainable management of Vertisols requires an integrated understanding of shrinkage mechanics, hydrological behaviour, structural deformation and electrochemical stability in order to minimize land degradation while utilizing the beneficial self-structuring properties associated with moderate shrink–swell dynamics.

## 5 | Effects of Soil Management Practices on Surface Cracking Parameters

Table 1 synthesizes published studies evaluating the effects of various management practices on soil cracking parameters across diverse climatic regions and cropping systems. Among the different management practices, organic residue retention and incorporation generally showed greater effectiveness in reducing crack development in Vertisols. In semi-arid Vertisols at Bhopal, incorporation of soybean straw (7.64 t $ha^{-1}$) reduced crack depth by 21.7%, compared with surface application (13.0%), indicating that subsurface organic inputs provide better buffering against shrinkage stresses. Similarly, Mohanty et al. (2006) reported substantial reductions in crack length (30.8%), crack depth (30.5%) and crack width (12.5%) under residue retention compared with residue removal. However, the influence of soil organic matter on structural stability and cracking behaviour in Vertisols appears to be complex and strongly dependent on clay mineralogy, soil chemical properties and moisture dynamics. Schweizer et al. (2022) observed that higher organic amendment application in a semiarid Vertisol did not significantly increase SOC or improve aggregate stability compared with integrated fertilization, suggesting that factors such as pH and exchangeable sodium may exert stronger control over soil structural behaviour than SOC alone. Nevertheless, Pierre et al. (2019) and Kishné et al. (2009) reported that increased organic matter may moderate shrink-swell intensity and improve structural resilience under suitable management conditions.

In contrast, mineral fertilization alone shows comparatively limited effectiveness; integrated nutrient management was generally more effective in reducing crack development. Application of 100% NPK resulted in modest reductions in crack width (3.80%–8.30%), crack depth (6.10%–6.90%) and crack volume (28.8%–32.1%), whereas combined FYM and NPK application substantially reduced crack width (11.5%–29.2%), crack depth (9.60%–26.6%) and crack volume (54.3%–69.2%). Although NPK fertilizers may indirectly improve aggregation and moisture regulation through enhanced root growth and rhizodeposition, these effects are generally slower and less pronounced than those achieved through integrated organic management practices.

Among chemical amendments, gypsum application was found to be highly effective, reducing crack depth and width by up to 40.6% and 46.2%, respectively. This effect is primarily attributed to calcium-induced flocculation of smectitic clays, which enhances structural stability and reduces shrinkage. Similarly, field and laboratory studies (Somasundaram et al. 2016) reported significant reductions in cracking parameters with organic amendments, with FYM application (20 Mg $ha^{-1}$) reducing crack volume by 18.0%–55.0%, more strongly associated with improved soil moisture retention. These findings reinforce the consensus that both organic and chemical amendments can effectively stabilize soil structure, although their mechanisms differ.

In contrast, certain management practices were found to increase cracking intensity. Cover crops, particularly those with deep rooting systems, enhanced soil water extraction and led to greater seasonal shrinkage, increasing crack depth, width and volume by 5.60%–51.5%, 30.8%–69.2% and 27.4%–90.2%, respectively. Similarly, no-tillage and reduced tillage increased cracking compared to conventional tillage, primarily due to surface compaction, residue-induced moisture stratification and deeper drying of the subsoil. Conversely, mouldboard ploughing reduced crack width (12.5%) and depth (6.70%) by loosening soil structure and disrupting continuous shrinkage planes. These contrasting responses indicate that the effectiveness of tillage practices depends on their influence on soil compaction, moisture distribution and structural continuity. Land configuration practices showed similarly contrasting effects. Conservation measures such as contour furrows and bench terraces reduced pot-hole volume by 73.4% and 48.5%, respectively, whereas uncultivated fields exhibited increases in pot-hole length (81.3%) and volume (27.4%), indicating unmanaged shrink-swell degradation.

A particularly strong contrast was observed between puddled rice systems and direct-seeded rice (DSR). Compared with DSR, puddling significantly increased crack width (37.0%–48.1%), depth (34.0%–34.7%) and total crack length (47.8%–49.8%), while crack surface area and volume increased dramatically by 98.7%–101% and 171%–192%, respectively (Bandyopadhyay et al. 2003). These findings are consistent with Mohanty et al. (2004), who attributed increased cracking to aggregate breakdown, clay dispersion, and formation of a compacted puddled layer with reduced macroporosity. During drying, this structurally weakened layer experiences greater shrinkage stresses, resulting in wider and deeper cracks. Although Mohanty et al. (2006) noted that puddling reduced crack initiation rates (18.0%–33.0%) but increased crack widening rates (56.0%–57.0%), suggesting a redistribution of shrinkage stresses. Overall, these studies consistently demonstrate that puddling amplifies shrink-swell expression and enhances crack connectivity, with important implications for preferential flow, water loss and soil structural degradation. Besides management practices, soil clay content and seasonal moisture regime also influence crack development in Vertisols. Under fallow conditions, the South Gezira site with higher clay content (57%–65%) and greater annual rainfall (430 mm) exhibited 108%, 133% and 828% higher crack depth, crack width and crack volume, respectively, than the North Gezira site with lower clay content (40%–45%) and rainfall (292 mm). The result suggests that the influence of soil moisture on cracking depends on its temporal dynamics, with seasonal wetting followed by drying intensifying shrink–swell behaviour



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**TABLE 1** | Soil cracking responses to different management practices under varied climatic and cropping conditions.

| Location | Soil type | Soil texture | Climate | Treatment/management component | Cracking parameter | Control | Treatment | % change due to treatment | References |
|---|---|---|---|---|---|---|---|---|---|
| Bhopal, India | Entic Chromusterts (Vertisol) | Silty clay (57.8% clay) | Semi-arid tropical | Soybean straw (surface application @ 7.64 t $ha^{-1}$) | Crack depth (mm) | 23.0 | 20.0 | 13.0% ↓ | Painuli et al. (2017) |
| | | | | Soybean straw (incorporation @ 7.64 t $ha^{-1}$) | Crack depth (mm) | 23.0 | 18.0 | 21.7% ↓ | |
| | | | | Soybean straw (incorporation @ 7.64 t $ha^{-1}$) | Crack width (mm) | 5.00 | 3.00 | 40.0% ↓ | |
| | | | | FYM @ 10 t $ha^{-1}$ | Crack depth (mm) | 32.0 | 25.0 | 21.9% ↓ | |
| | | | | FYM @ 10 t $ha^{-1}$ | Crack width (mm) | 13.0 | 9.00 | 30.8% ↓ | |
| | | | | Gypsum @ 10 t $ha^{-1}$ | Crack depth (mm) | 32.0 | 19.0 | 40.6% ↓ | |
| | | | | Gypsum @ 10 t $ha^{-1}$ | Crack width (mm) | 13.0 | 7.00 | 46.2% ↓ | |
| Kota, Rajasthan, India | Typic Chromusterts (Vertisol) | Clay loam | Semi-arid tropical | Wheat straw @ 2.5 g $kg^{-1}$ (lab incubation) | Crack volume | — | — | 74.0% ↓ | Somasundaram et al. (2016) |
| | | | | FYM @ 2.5 g $kg^{-1}$ (lab incubation) | Crack volume | — | — | 60.0% ↓ | |
| | | | | Fly ash @ 2.5 g $kg^{-1}$ (lab incubation) | Crack volume | — | — | 23.0% ↓ | |
| | | | | Wheat straw @ 20 Mg $ha^{-1}$ under grassland | Crack volume | — | — | 55.0% ↓ | |
| | | | | Soil amendments @ 20 Mg $ha^{-1}$ (cropland) | Crack volume | — | — | 18%–49% ↓ | |
| | | | | Soil amendments @ 20 Mg $ha^{-1}$ (grassland) | Crack volume | — | — | 18%–55% ↓ | |
| | | | | Soil amendments @ 20 Mg $ha^{-1}$ (fallow land) | Crack volume | — | — | 18%–43% ↓ | |

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**TABLE 1** | (Continued)

| Location | Soil type | Soil texture | Climate | Treatment/management component | Cracking parameter | Control | Treatment | % change due to treatment | References |
|---|---|---|---|---|---|---|---|---|---|
| Chambal region, Kota, Rajasthan, India | Typic Chromusterts (Vertisol) | Fine textured (>35% clay) | Semi-arid tropical | Contour furrow | Pot-hole length (m) | 0.32 | 0.24 | 25.0% ↓ | Somasundaram et al. (2011) |
| | | | | | Pot-hole width (m) | 0.27 | 0.20 | 25.9% ↓ | |
| | | | | | Pot-hole depth (m) | 0.39 | 0.17 | 56.4% ↓ | |
| | | | | | Pot-hole volume ($m^3 ha^{-1}$) | 450 | 119 | 73.4% ↓ | |
| | | | | Conservation bench terraces | Pot-hole length (m) | 0.32 | 0.18 | 43.8% ↓ | |
| | | | | | Pot-hole width (m) | 0.27 | 0.19 | 29.6% ↓ | |
| | | | | | Pot-hole depth (m) | 0.39 | 0.46 | 17.9% ↑ | |
| | | | | | Pot-hole volume ($m^3 ha^{-1}$) | 450 | 231 | 48.5% ↓ | |
| | | | | Residue incorporated | Pot-hole length (m) | 0.32 | 0.29 | 9.40% ↓ | |
| | | | | | Pot-hole width (m) | 0.27 | 0.17 | 37.0% ↓ | |
| | | | | | Pot-hole depth (m) | 0.39 | 0.22 | 43.6% ↓ | |
| | | | | | Pot-hole volume ($m^3 ha^{-1}$) | 450 | 142 | 68.3% ↓ | |
| | | | | Uncultivated field | Pot-hole length (m) | 0.32 | 0.58 | 81.3%↑ | |
| | | | | | Pot-hole width (m) | 0.27 | 0.41 | 51.9% ↑ | |
| | | | | | Pot-hole depth (m) | 0.39 | 0.31 | 20.5% ↓ | |
| | | | | | Pot-hole volume ($m^3 ha^{-1}$) | 450 | 573 | 27.4% ↑ | |
| Eastern China | Vertisol | Clayey | Subtropical monsoon | Woodchip biochar (WCB) @ 6.0% | Crack area density (%) | 13.2 | 11.4 | 13.6% ↓ | Zong et al. (2014) |
| | | | | Sludge biochar (WSB) @ 6.0% | Crack area density (%) | 13.4 | 11.1 | 17.2% ↓ | |
| | | | | Straw biochar (SB) @ 6.0% | Crack line density ($m\,m^{-2}$) | 36.5 | 29.5 | 19.2% ↓ | |
| | | | | Woodchip biochar (WCB) @ 6.0% | Crack line density ($m\,m^{-2}$) | 36.0 | 31.0 | 13.9% ↓ | |
| | | | | Sludge biochar (WSB) @ 6.0% | Crack line density ($m\,m^{-2}$) | 36.8 | 33.0 | 10.3% ↓ | |

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**TABLE 1** | (Continued)

| Location | Soil type | Soil texture | Climate | Treatment/management component | Cracking parameter | Control | Treatment | % change due to treatment | References |
|---|---|---|---|---|---|---|---|---|---|
| Córdoba, southern Spain | Vertisol, (Typic Haploxererts) | Clayey (69%–79% clay) | Mediterranean rainfed | No-tillage (NT) vs. Conventional tillage (CT) | Crack surface area ($m^2 m^{-2}$) | 4.99 | 4.01 | 19.6% ↓ | Lopez-Bellido et al. (2016) |
| | | | | NT vs. CT | Crack volume ($m^3 m^{-2}$) | 0.036 | 0.025 | 30.6% ↓ | |
| | | | | NT vs. CT (preceding crop: wheat) | Crack width (mm) | 25.0 | 24.0 | 4.00% ↓ | |
| | | | | NT vs. CT (fallow) | Crack width (mm) | 32.0 | 25.0 | 21.9% ↓ | |
| | | | | NT vs. CT (sunflower) | Crack width (mm) | 30.0 | 26.0 | 13.3% ↓ | |
| | | | | NT vs. CT (chickpea) | Crack width (mm) | 32.0 | 26.0 | 18.8% ↓ | |
| | | | | NT vs. CT (faba bean) | Crack width (mm) | 33.0 | 24.0 | 27.3% ↓ | |
| | | | | NT vs. CT (fallow) | Crack depth (mm) | 270 | 180 | 33.3% ↓ | |
| | | | | NT vs. CT (sunflower) | Crack depth (mm) | 260 | 230 | 11.5% ↓ | |
| | | | | NT vs. CT (chickpea) | Crack depth (mm) | 280 | 260 | 7.10% ↓ | |
| | | | | NT vs. CT (faba bean) | Crack depth (mm) | 250 | 230 | 8.00% ↓ | |

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**TABLE 1** | (Continued)

| Location | Soil type | Soil texture | Climate | Treatment/management component | Cracking parameter | Control | Treatment | % change due to treatment | References |
|---|---|---|---|---|---|---|---|---|---|
| Dharwad, Karnataka, India | Vertisol (Udic Haplustert) | Clay (~65% clay; montmorillonitic) | Dry sub-humid tropical | Chickpea cover crop | Crack width (mm) | 13.0 | 21.0 | 61.5% ↑ | Dasog and Shashidhara (1993) |
| | | | | Wheat cover crop | Crack width (mm) | 13.0 | 20.0 | 53.8% ↑ | |
| | | | | Sorghum cover crop | Crack width (mm) | 13.0 | 22.0 | 69.2% ↑ | |
| | | | | Sunflower cover crop | Crack width (mm) | 13.0 | 17.0 | 30.8% ↑ | |
| | | | | Safflower cover crop | Crack width (mm) | 13.0 | 22.0 | 69.2% ↑ | |
| | | | | Chickpea cover crop | Crack depth (mm) | 269 | 284 | 5.60% ↑ | |
| | | | | Wheat cover crop | Crack depth (mm) | 269 | 295 | 9.70% ↑ | |
| | | | | Sorghum cover crop | Crack depth (mm) | 269 | 324 | 20.4% ↑ | |
| | | | | Sunflower cover crop | Crack depth (mm) | 269 | 288 | 7.10% ↑ | |
| | | | | Safflower cover crop | Crack depth (mm) | 269 | 357 | 32.7% ↑ | |
| | | | | Chickpea cover crop | Crack length ($cm\,m^{-2}$) | 633 | 518 | 18.2% ↓ | |
| | | | | Wheat cover crop | Crack length ($cm\,m^{-2}$) | 633 | 605 | 4.40% ↓ | |
| | | | | Sorghum cover crop | Crack length ($cm\,m^{-2}$) | 633 | 574 | 9.30% ↓ | |
| | | | | Sunflower cover crop | Crack length ($cm\,m^{-2}$) | 633 | 600 | 5.20% ↓ | |
| | | | | Safflower cover crop | Crack length ($cm\,m^{-2}$) | 633 | 583 | 7.90% ↓ | |
| | | | | Chickpea cover crop | Crack volume ($m^3\,ha^{-1}$) | 234 | 310 | 32.5% ↑ | |
| | | | | Wheat cover crop | Crack volume ($m^3\,ha^{-1}$) | 234 | 393 | 67.9% ↑ | |
| | | | | Sorghum cover crop | Crack volume ($m^3\,ha^{-1}$) | 234 | 372 | 59.0% ↑ | |
| | | | | Sunflower cover crop | Crack volume ($m^3\,ha^{-1}$) | 234 | 445 | 90.2% ↑ | |
| | | | | Safflower cover crop | Crack volume ($m^3\,ha^{-1}$) | 234 | 353 | 50.9% ↑ | |
| Bhopal, Madhya Pradesh, India | Typic Haplustert (Vertisol) | Clay (~52% clay) | Hot sub-humid tropical | No tillage (NT) vs. conventional tillage (CT) | Crack width (mm) | 16.0 | 22.0 | 37.5% ↑ | Bandyopadhyay et al. (2003) |
| | | | | | Crack depth (mm) | 134 | 203 | 51.5% ↑ | |
| | | | | | Crack length ($mm\,m^{-2}$) | 5160 | 3220 | 37.6% ↓ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 1.39 | 1.31 | 5.80% ↓ | |
| | | | | | Crack volume ($\times 10^{-4}\,m^3\,m^{-2}$) | 55.5 | 72.2 | 30.0% ↑ | |

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**TABLE 1** | (Continued)

| Location | Soil type | Soil texture | Climate | Treatment/management component | Cracking parameter | Control | Treatment | % change due to treatment | References |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Reduced tillage (RT) vs. CT | Crack width (mm) | 16.0 | 25.0 | 56.3% ↑ | |
| | | | | | Crack depth (mm) | 134 | 185 | 38.1% ↑ | |
| | | | | | Crack length ($mm\,m^{-2}$) | 5160 | 3040 | 41.1% ↓ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 1.39 | 1.13 | 18.7% ↓ | |
| | | | | | Crack volume ($\times10^{-4}\,m^3\,m^{-2}$) | 55.5 | 71.0 | 27.8% ↑ | |
| | | | | Mould board ploughing (MBP) vs. CT | Crack width (mm) | 16.0 | 14.0 | 12.5% ↓ | |
| | | | | | Crack depth (mm) | 134 | 125 | 6.70% ↓ | |
| | | | | | Crack length ($mm\,m^{-2}$) | 5160 | 6490 | 25.8% ↑ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 1.39 | 1.63 | 17.3% ↑ | |
| | | | | | Crack volume ($\times10^{-4}\,m^3\,m^{-2}$) | 55.5 | 56.3 | 1.30% ↑ | |
| | | | | 100% NPK vs. 0% NPK (conventional tillage) | Crack width (mm) | 26.0 | 25.0 | 3.80% ↓ | |
| | | | | | Crack depth (mm) | 228 | 214 | 6.10% ↓ | |
| | | | | | Crack length ($mm\,m^{-2}$) | 5920 | 4770 | 19.4% ↓ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 2.70 | 2.04 | 24.4% ↓ | |
| | | | | | Crack volume ($\times10^{-4}\,m^3\,m^{-2}$) | 176 | 125 | 28.8% ↓ | |
| | | | | 100% NPK + FYM vs. 0% NPK (CT) | Crack width (mm) | 26.0 | 23.0 | 11.5% ↓ | |
| | | | | | Crack depth (mm) | 228 | 206 | 9.60% ↓ | |
| | | | | | Crack length ($mm\,m^{-2}$) | 5920 | 3470 | 41.4% ↓ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 2.70 | 1.43 | 47.0% ↓ | |
| | | | | | Crack volume ($\times10^{-4}\,m^3\,m^{-2}$) | 176 | 80.6 | 54.3% ↓ | |
| | | | | 100% NPK vs. 0% NPK (sub-soiling) | Crack width (mm) | 24.0 | 22.0 | 8.30% ↓ | |
| | | | | | Crack depth (mm) | 218 | 203 | 6.90% ↓ | |
| | | | | | Crack length ($mm\,m^{-2}$) | 5600 | 4550 | 18.8% ↓ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 2.44 | 1.85 | 24.2% ↓ | |
| | | | | | Crack volume ($\times10^{-4}\,m^3\,m^{-2}$) | 147 | 99.8 | 32.1% ↓ | |

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**TABLE 1** | (Continued)

| Location | Soil type | Soil texture | Climate | Treatment/management component | Cracking parameter | Control | Treatment | % change due to treatment | References |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 100% NPK + FYM vs. 0% NPK (sub-soiling) | Crack width (mm) | 24.0 | 17.0 | 29.2% ↓ | |
| | | | | | Crack depth (mm) | 218 | 160 | 26.6% ↓ | |
| | | | | | Crack length ($mm\,m^{-2}$) | 5600 | 3290 | 41.3% ↓ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 2.44 | 1.05 | 57.0% ↓ | |
| | | | | | Crack volume ($\times10^{-4}\,m^3\,m^{-2}$) | 147 | 45.3 | 69.2% ↓ | |
| | | | | Puddling (4 passes) vs. Direct seeding (DSR) | Crack width (mm) | 27.0 | 37.0 | 37.0% ↑ | |
| | | | | | Crack depth (mm) | 170 | 229 | 34.7% ↑ | |
| | | | | | Crack length ($m\,m^{-2}$) | 4.54 | 6.71 | 47.8% ↑ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 1.55 | 3.08 | 98.7% ↑ | |
| | | | | | Crack volume ($m^3\,m^{-2}\times10^{-4}$) | 105 | 284 | 171% ↑ | |
| | | | | Puddling (8 passes) vs. DSR | Crack width (mm) | 27.0 | 40.0 | 48.1% ↑ | |
| | | | | | Crack depth (mm) | 170 | 228 | 34.1% ↑ | |
| | | | | | Crack length ($m\,m^{-2}$) | 4.54 | 6.80 | 49.8% ↑ | |
| | | | | | Crack surface area ($m^2\,m^{-2}$) | 1.55 | 3.11 | 101% ↑ | |
| | | | | | Crack volume ($m^3\,m^{-2}\times10^{-4}$) | 105 | 306 | 192% ↑ | |
| Bhopal, Madhya Pradesh, India | Vertisol | Clayey | Sub-humid tropical | Puddling (4 passes) vs. Direct seeding (DSR) | Crack length rate ($cm\,day^{-1}$) | 14.5 | 10.1 | 30.5% ↓ | Mohanty et al. (2006) |
| | | | | | Crack width rate ($cm\,day^{-1}$) | 0.10 | 0.15 | 56.0% ↑ | |
| | | | | | Crack depth rate ($cm\,day^{-1}$) | 0.58 | 0.45 | 23.6% ↓ | |
| | | | | Puddling (8 passes) vs. DSR | Crack length rate ($cm\,day^{-1}$) | 14.5 | 9.80 | 32.4% ↓ | |
| | | | | | Crack width rate ($cm\,day^{-1}$) | 0.10 | 0.15 | 57.0% ↑ | |
| | | | | | Crack depth rate ($cm\,day^{-1}$) | 0.58 | 0.48 | 18.0% ↓ | |
| | | | | Residue retained vs. residue removed | Crack length rate ($cm\,day^{-1}$) | 13.6 | 9.42 | 30.8% ↓ | |
| | | | | | Crack width rate ($cm\,day^{-1}$) | 0.13 | 0.11 | 12.5% ↓ | |
| | | | | | Crack depth rate ($cm\,day^{-1}$) | 0.59 | 0.41 | 30.5% ↓ | |

(Continues)



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**TABLE 1** | (Continued)

| Location | Soil type | Soil texture | Climate | Treatment/management component | Cracking parameter | Control | Treatment | % change due to treatment | References |
|---|---|---|---|---|---|---|---|---|---|
| Gezira, Sudan | Vertisol | Clayey | Semi-arid | Wheat crop under relatively lower clay content and intermediate rainfall/ Wheat crop vs. fallow land | Crack depth (mm) | 480 | 440 | 8.30%↓ | Elias et al. (2001) |
| | | | | | Crack width (mm) | 49.0 | 35.0 | 28.6%↓ | |
| | | | | | Crack volume ($m^3 ha^{-1}$) | 650 | 230 | 64.6%↓ | |
| | | | | Low clay (40%–45%) vs. high clay (57%–65%) and wetter moisture regime in fallow land | Crack depth (mm) | 230 | 480 | 108%↑ | |
| | | | | | Crack width (mm) | 21.0 | 49.0 | 133%↑ | |
| | | | | | Crack volume ($m^3 ha^{-1}$) | 7.00 | 65.0 | 828%↑ | |

and promoting more extensive crack formation in Vertisols (Elias et al. 2001). In Vertisols, management practices influence cracking behaviour through interconnected effects on soil structure, shrinkage characteristics, moisture dynamics and hydraulic behaviour. For conceptual clarity, the practices reviewed here are grouped according to their predominant mechanism of action into: (i) agronomic, biological and soil structural management practices and (ii) hydrological and land-configuration practices (Figure 5).

**(i) Agronomic, biological and soil structural management practices:** These practices primarily influence crack development by modifying soil aggregation, pore continuity, shrinkage stress and soil mechanical behaviour.

Mulching influences shrinkage and cracking in Vertisols primarily by moderating soil moisture fluctuations, reducing evaporation losses and lowering tensile stress development during drying (Rao et al. 1977; Mulumba and Lal 2008), yet the magnitude of this effect varies considerably across studies, and this variability is not incidental but reflects real differences in climate, mulch material and management context rather than inconsistent methodology. In semi-arid regions, where evaporative demand is high and residue decomposes slowly, surface mulch acts as a protective barrier against direct solar radiation and wind exposure, thereby slowing soil water loss and maintaining a more uniform moisture distribution within the soil profile (Gicheru 1994; Nalatwadmath et al. 2006; Meena et al. 2011). In humid environments, by contrast, frequent rainfall already maintains higher baseline soil moisture, so the marginal moisture benefits that mulch can add are smaller, and faster residue decomposition further shortens its period of effectiveness, which explains why studies from wetter climates tend to report modest or inconsistent crack-reduction effects (Table 2). Mulching material and application rate are important factors, as reported by Somasundaram et al. (2016) incubation and field trials on Vertisols show that wheat straw outperforms sawdust, FYM, gypsum and fly ash in reducing crack volume, with effectiveness increasing at higher application rates, indicating that comparisons using different residue types or low-rate applications will understate mulching's true potential. Soil texture and clay mineralogy add another layer of variation since crack width and depth are governed as much by shrink-swell potential, clay content and confining pressure as by moisture status alone (Painuli et al. 2017); thick organic mulch is therefore likely to succeed most clearly in high shrink–swell Vertisols, where self-mulching and cracking are pronounced (Somasundaram, Lal, et al. 2018), and to show weaker, harder-to-detect effects in less expansive or better-drained soils. Traditional dust mulching after harvest in central India illustrates another variable: practice type and timing, since shallow post-harvest tillage suppresses early crack initiation even without organic residue (Somasundaram, Singh, et al. 2018), suggesting that studies comparing organic mulch against tillage-based mulching are not measuring the same mechanism and should not be expected to agree. Finally, mulching's benefits extend beyond direct crack suppression to improved aggregate stability, infiltration and biological activity (Table 3) (Rathore et al. 1998), and because cracking itself aids natural structural regeneration in Vertisols through wetting-drying cycles (Pillai and McGarry 1999), the practice is most likely to succeed where residues or growing crops are present



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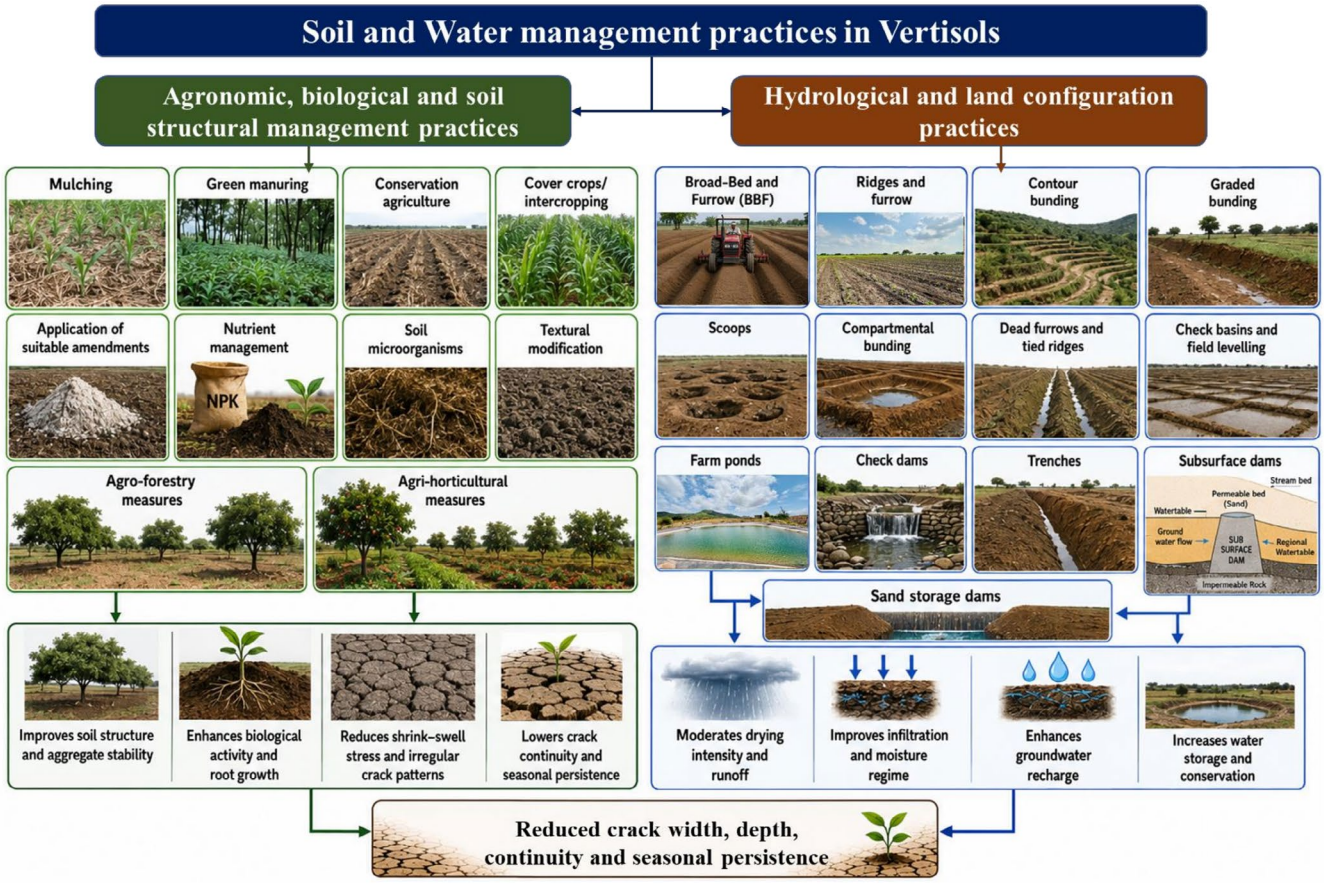



**FIGURE 5** | Conceptual framework showing the classification of soil and water management and their pathways of influence on soil cracking and shrinkage in Vertisols. The visual presentation of the figure was enhanced using Canva AI (Canva), an AI-assisted design tool.

to convert that cracking into productive aggregation (Pillai and McGarry 1999; Bravo-Garza et al. 2009) and most likely to underperform where cracks form on bare, saturated soil and instead promote runoff and soil loss (Carroll et al. 1997). Taken together, these findings suggest mulching effectiveness in Vertisols is not a fixed property of the practice but an outcome of the interaction between climate aridity, residue type and rate, soil shrink-swell capacity and management timing—a distinction future comparative studies should control for explicitly.

The green manuring and legume-based biomass incorporation improve soil aggregation, pore continuity and biological activity, thereby enhancing the structural resilience of Vertisols during repeated wetting-drying cycles. The incorporation of green biomass increases soil organic matter and promotes the formation of organic binding agents that stabilize aggregates and improve moisture retention (Munirathnam et al. 2026). In Vertisol-dominated regions of India, approximately 14.0 million ha remain fallow during the kharif season because of erratic monsoon rainfall and unfavourable soil physical conditions (Rao et al. 2015; Directorate of Economics and Statistics 2024). Under such circumstances, the cultivation of green manure crops such as sunhemp, dhaincha, cowpea and *Sesbania* sp. during the pre-monsoon period improves infiltration, suppresses weeds, reduces runoff, and enhances residual moisture availability for succeeding crops (Page et al. 2019; Dengia et al. 2024). Becker et al. (1995) further reported that *Sesbania* can fix approximately 75.0–80.0 kg N $ha^{-1}$ within 50–60 days, thereby improving nutrient cycling and soil biological activity. Regassa and Elias (2022) reported that the integration of legume species such as vetch and clover into Vertisol-based cropping systems improves farm-level resource-use efficiency while simultaneously protecting the soil against runoff and erosion through enhanced vegetative cover and biomass production. However, the reported effectiveness of green manuring varies considerably among studies because of differences in climate, residue characteristics, soil condition and management practices. Greater improvements are generally reported in semi-arid regions, where slower residue decomposition allows organic matter to persist longer, resulting in sustained improvements in soil structure and moisture retention (Srivastava et al. 1989; Dasog and Shashidhara 1993; Somasundaram et al. 2016). In contrast, studies from humid regions often report smaller reductions in cracking because rapid decomposition shortens the duration of soil cover and naturally higher soil moisture already limits crack development. The response also depends on biomass quality and quantity, as high-biomass leguminous residues with low C:N ratios generally improve aggregation more effectively than low-biomass residues (Mandal et al. 2003).

Building upon these biologically driven approaches, the effects of conservation agriculture (CA) on Vertisol structure vary among studies due to differences in initial soil condition, duration of adoption, climate and residue management rather than differences in the underlying mechanism (Table 3). The long-term studies generally report greater improvements in



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**TABLE 2** | Effect of different cracking management practices on runoff, soil loss and water use efficiency in Vertisols.

| Study location | Moisture-management practices | Soil type | Reduction in runoff (%) | Reduction in soil loss (%) | Increase in water use efficiency (%) | Reference(s) |
|---|---|---|---|---|---|---|
| Agronomic, biological and soil structural management practices | | | | | | |
| Ballari (Karnataka) | Mulching with sorghum and *dolichos* | Vertisols | 16.0% | 31.6% | 21.5% | Nalatwadmath et al. (2006) |
| Bhopal, Madhya Pradesh, India | Soybean (sole crop) | Vertisol (Typic Haplustert) | 34.1% | 43.0% | 12.0% | Singh et al. (2020) |
| | Maize (sole crop) | | 21.3% | 29.0% | 7.00% | |
| | Pigeon pea (sole crop) | | 15.1% | 22.0% | 4.00% | |
| | Soybean + maize (1:1) | | 31.5%–36.2% | 42.0%–45.0% | 11.0% | |
| | Soybean + pigeon pea (2:1) | | 34.0%–36.0% | 45.0%–51.0% | 12.0% | |
| | Maize + pigeon pea (1:1) | | 27.0%–32.0% | 37.0%–45.0% | 10.0% | |
| ICAR-IISS, Bhopal, MP, India | $Mulch_{50}$ (50% surface cover) | Vertisol | 43.3% | 61.1% | — | Painuli et al. (2017) |
| | $Mulch_{100}$ (100% surface cover) | | 63.3% | 88.9% | — | |
| | $Mulch_{200}$ (200% surface cover) | | 73.3% | 94.4% | — | |
| | FYM @ 16 t $ha^{-1}$ | | 28.0%–41.0% | — | — | |
| | Gypsum @ 4 t $ha^{-1}$ | | 47.0%–84.0% | — | — | |
| | Gypsum @ 16 t $ha^{-1}$ | | 63.0%–89.0% | — | — | |

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**TABLE 2** | (Continued)

| Study location | Moisture-management practices | Soil type | Reduction in runoff (%) | Reduction in soil loss (%) | Increase in water use efficiency (%) | Reference(s) |
|---|---|---|---|---|---|---|
| Kota, Rajasthan, India | Recommended dose of NPK (RNPK) | Vertisol (silty clay loam) | 10.3% | 26.1% | — | Rashmi et al. (2025) |
| | RNPK + Gypsum (2.5 t ha$^{-1}$) | | 28.7% | 43.5% | — | |
| | RNPK + FYM (10 t ha$^{-1}$) | | 20.5% | 39.1% | — | |
| | RNPK + Crop residue (1.5–3.0 t ha$^{-1}$) | | 22.9% | 47.8% | — | |
| | RNPK + Gypsum + FYM | | 44.1% | 54.3% | — | |
| | RNPK + Gypsum + Crop residue | | 38.6% | 50.0% | — | |
| | RNPK + Gypsum + Crop residue + FYM | | 40.6% | 52.2% | — | |
| Kota, Rajasthan (Chambal ravines), India | Agri-horticulture (Soybean + *Manilkara achras*) | Vertisols (Typic Haplustert) | 1.40% (increased) | 16.6% (increased) | — | Meena et al. (2023) |
| | Horti-pasture (*Emblica officinalis* + *Cenchrus ciliaris*) | | 7.50% | 13.9% | — | |
| | Pasture (*Cenchrus ciliaris*) | | 6.30% | 18.0% | — | |
| | Silviculture (*Acacia nilotica*) | | 16.4% (increased) | 37.3% (increased) | — | |
| | Silvi-pasture (*Acacia nilotica* + *Cenchrus ciliaris*) | | 4.00% (increased) | 5.30% | — | |

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**TABLE 2** | (Continued)

| Study location | Moisture-management practices | Soil type | Reduction in runoff (%) | Reduction in soil loss (%) | Increase in water use efficiency (%) | Reference(s) |
|---|---|---|---|---|---|---|
| Queensland, Australia (Greenmount) | Stubble incorporated | Vertisols (Udic Pellustert) | 28.1% | 70.7% | — | Freebairn and Wockner (1986) |
| | Stubble mulch | | 34.6% | 91.3% | — | |
| | Zero tillage | | 18.5% | 96.6% | — | |
| | Summer crop | | 38.8% | 63.4% | — | |
| Queensland, Australia (Greenwood) | Stubble incorporated | Vertisols (Udic Chromustert) | 20.9% | 75.3% | — | |
| | Stubble mulch | | 29.6% | 87.7% | — | |
| | Zero tillage | | 25.7% | 94.3% | — | |
| | Summer crop | | 22.5% | 37.3% | — | |
| Sinana, Bale highlands, SE Ethiopia | Straw mulch @ 2 t $ha^{-1}$ | Vertisol (clayey) | 84.7% | 94.8% | 19.0% | Birru et al. (2012) |
| | Straw mulch @ 4 t $ha^{-1}$ | | 96.7% | 98.9% | 21.7% | |
| | Straw mulch @ 6 t $ha^{-1}$ | | 98.3% | 99.7% | 22.1% | |
| | FYM @ 2 t $ha^{-1}$ | | 12.2% | 32.2% | 2.53% | |
| | FYM @ 4 t $ha^{-1}$ | | 3.30% | 2.30% | 0.74% | |
| | FYM @ 6 t $ha^{-1}$ | | 11.3% | 3.30% | 2.70% | |
| Kota, southeastern Rajasthan, India | Crop residue (wheat straw) application | Vertisol (Typic Chromusterts, clay loam) | — | — | 13.5% | Somasundaram, Lal, et al. (2018), Somasundaram, Singh, et al. (2018) |
| | FYM @ 5 Mg $ha^{-1}$ | | — | — | 8.12% | |
| | Fly ash @ 10 Mg $ha^{-1}$ | | — | — | 12.5% | |
| | Gypsum (100% GR) | | — | — | 5.69% | |
| ICRISAT, Patancheru, India | Conservation agriculture (minimum tillage + crop residue retention under BBF system) | Vertisol (medium black clayey soil) | 28.0%–30.0% | — | 10.0%–40.0% | Patil et al. (2016) |
| Hydrological and land-configuration practices | | | | | | |
| Bhopal, India | BBF and FOG | Vertisol (Typic Haplustert) | 24.2%–42.5% | 31.1%–55.2% | 11%–15% | Mandal et al. (2013) |
| Bhopal, India | BBF and cropping system | Vertisols | — | — | 22.5%–41.5% | Singh et al. (2011) |

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**TABLE 2** | (Continued)

| Study location | Moisture-management practices | Soil type | Reduction in runoff (%) | Reduction in soil loss (%) | Increase in water use efficiency (%) | Reference(s) |
|---|---|---|---|---|---|---|
| Parbhani, Maharashtra, India | Broad Bed and Furrow (BBF) | Vertisol | — | — | 26.3% | Asewar et al. (2017) |
| | Ridges and Furrow | Vertisol | — | — | 14.0% | |
| Rajkot, Gujarat, India | Broad bed (60 cm) + furrow (30 cm), 2 rows (BBF) | Medium black soil (Vertisol, clayey) | 6.44% | 11.7% | 3.40% | Vekariya et al. (2015) |
| | Broad bed (90 cm) + furrow (45 cm), 3 rows (BBF) | | 11.3% | 17.0% | 5.40% | |
| | Broad bed (120 cm) + furrow (60 cm), 4 rows (BBF) | | 6.49% | 10.9% | 1.40% | |
| May Zegzeg catchment, Tigray, Northern Ethiopia | Derdero+ (DER+) – permanent raised beds + ~30% crop residue retention (CA) vs. CT | Vertisol | 55.0% | 75.0% | 20.0% | Araya et al. (2012) |
| | Terwah+ (TER+) – contour furrows (~1.5 m spacing) + ~30% crop residue retention (CA) vs. CT | Vertisol | 24.0% | 42.0% | 11.0% | |
| Ethiopian highlands (various watersheds), Ethiopia | Tied ridging | Vertisols | 55.8% | 62.0% | — | Adimassu et al. (2017) |
| | Stone bunds | Vertisols | 34.4% | 38.0% | — | |
| | Level soil bunds | Vertisols | 60.1% | 88.0% | — | |
| | Mulching (crop residue) | Vertisols | 75.5% | 98.0% | — | |
| Ethiopia | Conservation agriculture (Derdero: permanent beds + residue retention) | Vertisols | 37.6% | 59.3% | — | Lanckriet et al. (2012) |

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**TABLE 2** | (Continued)

| Study location | Moisture-management practices | Soil type | Reduction in runoff (%) | Reduction in soil loss (%) | Increase in water use efficiency (%) | Reference(s) |
|---|---|---|---|---|---|---|
| Ethiopia | Different CSB-SWC techniques (*Fanya juu*, soil bunds, stone bunds, bench terraces, vegetative barriers, and tied-ridges) | Vertisols | 13.0%–71.0% | 39.0%–83.0% | — | Wolka et al. (2018) |
| Sehore (MP), Kothapally (TG), Bhopal (MP) | Farm ponds (rainwater harvesting) | Vertisols | 20.0%–30.0% | — | — | Wani et al. (2003) |
| Ballari (Karnataka) | Graded bunding (bunds with 0.6 sq.m. cross section spaced at 0.75 m vertical interval) | Vertisols | 74.2% | 85.0% | — | Rao et al. (1998) |
| Ballari (Karnataka) | Compartment bunding | Vertisols | — | — | 15.5% | Patil et al. (2016) |
| Bhopal, India | Broad-Bed & Furrow (BBF) vs. flat on grade (FOG) | Vertisol (Typic Haplustert) | 24.0%–32.0% | 31.0%–55.0% | — | Hati et al. (2013) |
| Hyderabad, India | Raised beds/BBF | Vertisols | — | 30.0%–60.0% | — | Garg et al. (2022) |



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**TABLE 3** | Soil and water conservation practices for mitigating crack development and enhancing resilience of Vertisols.

| Soil and water conservation practices | Sub-practices | Primary mechanism | Effect on cracking | Additional hydrological/ structural effect | Limitation | References |
|---|---|---|---|---|---|---|
| Agronomic, biological and soil structural management practices | Mulching, residue retention and alternative organic surface covers | Moderate soil moisture fluctuations and reduce evaporation losses through surface protection | Mitigate shrinkage intensity and crack connectivity | Improve soil moisture conservation, aggregation and erosion control | Effectiveness depends on mulch type, application quantity, residue availability, decomposition dynamics and climatic conditions | Rao et al. (1977), Rathore et al. (1998), Birru et al. (2012), Somasundaram et al. (2016), Somasundaram, Singh, et al. (2018) |
| | Green manuring and legume-based biomass incorporation | Enhance aggregation, biological activity and moisture retention through biomass addition | Reduce shrinkage stress and crack persistence | Improve soil fertility, infiltration, moisture conservation and erosion control | Performance depends on biomass production, rainfall variability and timing of incorporation | Becker et al. (1995), Mandal et al. (2003), Blaise (2011), Page et al. (2019), Dengia et al. (2024), Munirathnam et al. (2026) |
| | Conservation agriculture and reduced soil disturbance practices | Minimize soil disturbance and maintain residue cover to stabilize soil structure | Mitigate shrink–swell cracking and structural degradation | Improve soil hydraulic properties, aggregation, soil carbon and erosion resistance | Effectiveness depends on residue retention and availability, soil type, climatic region and long-term management | Bandyopadhyay et al. (2003), Erkossa et al. (2005), Tennakoon and Hulugalle (2006), McHugh et al. (2009), Boulal et al. (2011), Palm et al. (2014), Lenka et al. (2014), Patil et al. (2016), Lopez-Bellido et al. (2016), Botta et al. (2022), Somasundaram et al. (2022) |
| | Cover crops, intercropping and crop diversification systems | Maintain continuous soil cover and active root networks | Moderate drying-induced cracking and moisture stress | Improve infiltration, nutrient cycling, porosity and resilience to climatic variability | Benefits vary with species selection, biomass production and rainfall conditions | Nyalemegbe et al. (2011), Mandal et al. (2013), Singh et al. (2014), Singh et al. (2020), Zhang et al. (2022), Reddy et al. (2025) |
| | Gypsum application and sodicity amelioration | Replace exchangeable $Na^+$ with $Ca^{2+}$ to promote flocculation and aggregation | Reduce crack intensity and shrink–swell stress | Improve porosity, hydraulic conductivity, moisture retention and erosion resistance | Effectiveness depends on sodicity level, gypsum rate and leaching conditions | Hulugalle, McCorkell, et al. (2010), Hulugalle, Weaver, and Finlay (2010), Somasundaram et al. (2016), Painuli et al. (2017), Shah et al. (2017), Rashmi et al. (2018), Rahina et al. (2020), Bello et al. (2021), Dong et al. (2022), Niaz et al. (2023), Rashmi et al. (2024), Rashmi et al. (2025) |

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**TABLE 3** | (Continued)

| Soil and water conservation practices | Sub-practices | Primary mechanism | Effect on cracking | Additional hydrological/ structural effect | Limitation | References |
|---|---|---|---|---|---|---|
| | Integrated nutrient and organic matter management | Improve soil organic matter dynamics and microbial activity | Reduce crack development and structural degradation | Enhance aggregation, water retention, nutrient cycling and soil resilience | Requires balanced nutrient supply and long-term organic matter inputs | Bandyopadhyay et al. (2003), Wani et al. (2003), Hati et al. (2006), Hati, Swarup, et al. (2007), Bandyopadhyay et al. (2010), Murphy (2015), Somasundaram et al. (2016), Painuli et al. (2017), Zhao et al. (2020), Al-Nasser and Al-Khalid (2020) |
| | Soil microorganisms and biologically mediated aggregation | Strengthen aggregate cohesion through fungal, microbial and enzymatic activity | Improve resistance to shrink–swell cracking | Enhance pore stability, infiltration, nutrient cycling and soil resilience | Biological responses depend on soil moisture, organic matter and climatic conditions | Bearden and Petersen (2000), Miller and Jastrow (2000), Neergaard Bearden (2001), Hati, Biswas, et al. (2007), Rahman et al. (2017), DeCarlo and Caylor (2019) |
| | Texture modification through sand incorporation | Reduce dominance of expansive clay fractions and improve pore distribution and soil structure | Reduce crack propagation and volumetric deformation | Improve permeability, aggregation and structural stability | Excessive sand may reduce water-holding capacity and increase implementation cost | Dulur et al. (2015), Sabari Ramesh and Thyagaraj (2020), Mindari et al. (2023), Alnmr and Ray (2024), Qusai et al. (2025) |
| | Agroforestry and agri-horticultural systems | Deep perennial roots and residue inputs reinforce soil structure | Reduce crack depth and continuity | Improve infiltration, soil carbon accumulation and long-term resilience | Benefits depend on species selection, rooting pattern and establishment period | Ramachandra (2010), Fadl (2012), Mohanty et al. (2012), Shabtai et al. (2014), Fahad et al. (2022), Meena et al. (2023), Wang et al. (2024), Pan et al. (2025), Pulido-Esquivel et al. (2025), Ramesha et al. (2025) |
| Hydrological and land-configuration practices | Broad-Bed and Furrow (BBF) system | Regulate runoff and conserve moisture through raised beds and furrows | Reduce desiccation stress and deep crack formation | Improve drainage, infiltration, aeration and water-use efficiency | Requires proper bed dimensions, slope management and maintenance | Pathak et al. (2011), Araya et al. (2011), Mandal et al. (2013), Erkossa et al. (2014), Asewar et al. (2017), Anantha et al. (2021), Garg et al. (2022), Kumawat et al. (2025) |
| | Ridge and furrow system | Enhance infiltration and temporary water storage within furrows | Moderate moisture stress and shrinkage cracking | Improve soil water storage, aeration and crop productivity | Suitability varies with rainfall regime and soil moisture conditions | Erkossa et al. (2005), Pathak et al. (2011), Patil et al. (2011), Ren et al. (2017), Rao et al. (2022), Kadry and El-Gawad (2023) |

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**TABLE 3** | (Continued)

| Soil and water conservation practices | Sub-practices | Primary mechanism | Effect on cracking | Additional hydrological/ structural effect | Limitation | References |
|---|---|---|---|---|---|---|
| | Contour bunding and contour-oriented ridges | Slow runoff and conserve rainwater along contour lines | Reduce moisture fluctuations and crack development | Improve moisture conservation, nutrient retention and erosion control | Effectiveness depends on slope, rainfall distribution and bund maintenance | Birhanu et al. (2022), Rao et al. (2022), Rao et al. (2024), Singh et al. (2025) |
| | Graded bunding | Safely drain excess runoff while reducing runoff velocity | Moderate desiccation stress and shrinkage cracking | Prevent waterlogging and reduce runoff and soil loss | Requires appropriate slope design and maintenance of waterways | Pathak et al. (2011), Rao et al. (2022) |
| | Scoops or pitting technique | Harvest runoff within small micro catchments to improve infiltration | Reduce wetting–drying intensity and surface cracking | Improve soil moisture availability and crop establishment | Benefits vary with rainfall intensity, soil cover and local conditions | Pathak et al. (2011), Kimaru-Muchai et al. (2021) |
| | Compartmental bunding | Retain rainwater within field compartments for gradual infiltration | Delay deep crack formation by maintaining uniform moisture | Improve water-use efficiency, root growth and crop productivity | Depends on rainfall pattern, bund maintenance and field slope | Patil and Sheelavantar (2004), Patil et al. (2015), Patil et al. (2016), Rao et al. (2022) |
| | Dead furrows and tied ridges | Temporarily store rainwater within in situ micro basins | Reduce desiccation stress and shrinkage cracking | Improve moisture availability, nutrient retention and runoff control | Performance depends on rainfall distribution and field management | Gebrekidan (2003) |
| | Subsurface dams and subsurface barriers | Restrict lateral subsurface water movement and conserve groundwater | Reduce deep drying fronts and seasonal crack persistence | Enhance groundwater recharge and long-term soil moisture stability | Requires suitable geological conditions and higher construction cost | Onder and Yilmaz (2005), Raju et al. (2006), Frankl et al. (2011), Ishida et al. (2011), Glendenning et al. (2012), Dortaj et al. (2020), Ebrahimi et al. (2021), Shen et al. (2025) |
| | Sand storage dams | Store water within sandy streambed deposits to enhance recharge | Indirectly reduce seasonal cracking through improved moisture availability | Improve drought resilience and supplemental irrigation potential | Limited to suitable sandy alluvial catchments | Hanson and Nilsson (1986), Aerts et al. (2007), Lasage et al. (2015), Neufeld et al. (2021), Lazurko et al. (2024) |

(Continues)



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**TABLE 3** | (Continued)

| Soil and water conservation practices | Sub-practices | Primary mechanism | Effect on cracking | Additional hydrological/ structural effect | Limitation | References |
|---|---|---|---|---|---|---|
| | Farm ponds and rainwater harvesting structures | Store surplus runoff for supplemental irrigation | Reduce severity and persistence of shrinkage cracks | Improve groundwater recharge, irrigation resilience and drought adaptation | Effectiveness depends on rainfall, storage capacity and maintenance | Sharma et al. (2004), Garg et al. (2020), Vico et al. (2020), Anantha et al. (2021), Neswati et al. (2023), Baruah and Hazarika (2025) |
| | Check dams and water-harvesting structures | Store runoff and enhance infiltration and groundwater recharge | Indirectly reduce cracking through improved soil moisture availability | Reduce soil loss, improve irrigation security and enhance productivity | Depends on catchment conditions, sedimentation and long-term maintenance | Garg et al. (2012), Polyakov et al. (2014), Piton et al. (2017), Djuma et al. (2017), Dashora et al. (2018) |
| | Conservation trenching | Intercept runoff and improve in situ moisture conservation | Reduce desiccation stress and shrinkage cracking indirectly | Improve vegetation growth, residual moisture utilization and erosion control | Effectiveness depends on slope, rainfall, trench design and vegetation suitability | Beckers et al. (2013), Kurothe et al. (2014), Alem et al. (2020), Shaikh et al. (2021), Ali et al. (2024) |

soil structure and carbon storage because aggregate stabilization and pore development under minimum tillage occur gradually with continuous residue retention (Hati et al. 2015). The CA is widely recognized as an effective strategy for improving soil moisture retention, structural stability and long-term sustainability of Vertisols (Palm et al. 2014; Patil et al. 2016). The core principles of CA-minimum soil disturbance, permanent soil cover and crop diversification-operate synergistically to regulate soil moisture dynamics, improve aggregation and enhance soil physical functioning (Jat et al. 2020; FAO 2022). Reduced tillage preserves soil structure and aggregate continuity, thereby improving infiltration and reducing rapid surface drying (McHugh et al. 2009; Boulal et al. 2011). Consequently, conservation-based systems moderate shrinkage intensity and reduce the formation of deep and wide cracks commonly observed in Vertisols during prolonged dry periods (Lopez-Bellido et al. 2016; Somasundaram et al. 2017). The CA improves soil physical properties, reduces soil loss and enhances soil organic carbon accumulation, thereby strengthening soil resilience and productivity (Nyakatawa et al. 2001). However, the differences in crop diversification and residue quantity also contribute to contrasting results, as inadequate surface residue limits the ability of CA to buffer shrink-swell cycles. The CA is particularly effective on Vertisols in semi-arid and subtropical regions because minimum tillage allows the natural wetting-drying cycle to consolidate rather than disrupt soil structure. On Vertisol (Vertosol) soils in north-western New South Wales, Australia, Tennakoon and Hulugalle (2006) found that irrigation efficiency and infiltration increased by 10%–35% as the season progressed under minimum tillage, attributing this improvement to intense wetting-drying cycles that strengthened furrow soil structure when left undisturbed by tillage. The same study reported that average seasonal evapotranspiration was higher under minimum tillage than conventional tillage, with plant-available water in minimum-tilled cotton increased by 18 mm over conventionally tilled systems. These structural benefits proved durable over the long term: across multiple studies on Vertosols in this region, minimum tillage, permanent bed maintenance, wheat rotation and stubble retention were consistently found to improve soil structural condition and subsequent water-use efficiency. Page et al. (2020) extended this evidence over five decades on a semi-arid subtropical Vertisol, showing that no-tillage combined with stubble retention and adequate nitrogen fertilization increased soil organic carbon and nitrogen stocks in the surface layer by 12.8% and 31.7%, respectively, relative to conventional tillage with stubble burning and no fertilization. The consistency of these gains across both short-term water-use studies and multi-decade carbon studies indicates that semi-arid conditions provide the specific combination required for CA to succeed on Vertisols: rainfall sufficient to trigger regular shrink–swell cycling, but not so continuous or intense as to prevent residues from persisting long enough to buffer that cycling and prevent surface desiccation.

An important component of conservation-oriented management is controlled traffic farming (CTF), which further improves the physical condition of Vertisols by minimizing wheel-induced compaction (Radford et al. 2007). Restricting machinery movement to designated lanes preserves pore continuity, improves aggregation, and enhances infiltration and root proliferation in non-trafficked zones (McHugh et al. 2009; Botta et al. 2022).



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These improvements promote more uniform soil moisture distribution and reduce shrink-swell stresses under repeated wetting-drying cycles. Quantitative evidence also highlights the effectiveness of CA practices in reducing crack development. Bandyopadhyay et al. (2003) reported that subsoiling reduced crack surface area, crack volume, crack width and crack depth by 11.9%, 23.7%, 12.5% and 10.2%, respectively, compared with conventional tillage under a soybean-linseed rotation. Similarly, Lopez-Bellido et al. (2016) observed reductions of approximately 19.6% in crack surface area and 30.6% in crack volume under long-term no-tillage systems. Besides improving structural stability, conservation agriculture also substantially improves hydrological performance by enhancing infiltration and reducing runoff and erosion losses. Patil et al. (2016) reported that minimum tillage combined with residue retention reduced runoff by 28.0%–30.0% and increased water-use efficiency by 10.0%–40.0% compared with conventional management. Likewise, no-tillage systems with high residue cover (> 90%) reduced runoff by 42.0% and soil loss by 77.0% in Vertisols relative to chisel tillage (Potter et al. 1995; Pantone et al. 1996). In Ethiopian Vertisols, reduced tillage lowered runoff and soil loss by 5.30% and 13.8%, respectively, compared with broad-bed and furrow systems (Erkossa et al. 2005). Moreover, zero-tillage with diversified cropping systems maintains higher ground cover and significantly reduces runoff and sediment loss in Vertisols under wheat-, sorghum- and sunflower-based systems (Freebairn and Wockner 1986; Richardson and King 1995). Despite these benefits, poorly managed no-tillage systems may sometimes increase cracking because of insufficient residue cover or localized surface compaction (Lenka et al. 2014). Nevertheless, long-term studies consistently demonstrate that CA improves soil aggregation (20.8%–26.0%) and soil organic carbon (12.9%–19.4%) in Vertisols (Somasundaram et al. 2022). Overall, by improving infiltration, conserving soil moisture, reducing runoff and erosion, and moderating shrink-swell stresses, conservation agriculture enhances the long-term structural resilience and sustainability of Vertisol-dominated agroecosystems.

In addition to conservation agriculture, cover crops and intercropping systems improve soil moisture regimes through enhanced infiltration, increased surface roughness and continuous soil cover, thereby reducing runoff velocity and erosion-induced soil loss (Singh et al. 2014). Maintaining at least 30% ground cover is considered critical for erosion control. Supporting this, Van Eerd et al. (2023) reported that 1.0–2.0 $Mg\,ha^{-1}$ dry biomass can provide 50.0%–70.0% soil coverage and reduce erosion by 80.0%–97.0% compared with bare soil. Similarly, soybean-based intercropping systems such as soybean + maize and soybean + pigeon pea significantly reduced runoff, soil and nutrient losses in Vertisols relative to cultivated fallow (Singh et al. 2020). Cover crops including *Mucuna pruriens*, *Mimosa invisa* and *Stylosanthes guianensis* also improved soil porosity, infiltration and moisture storage compared with natural fallow conditions (Nyalemegbe et al. 2011). According to Sujathamma and Nedunchezhiyan (2024) sorghum-blackgram intercropping (2:2 row ratio) proved to be the most productive and profitable cropping system for rainfed Vertisols under semi-arid conditions. The extensive root systems of cover crops create continuous biopores that improve water movement and porosity, whereas decomposed residues enhance soil organic matter and aggregate stability (Zhang et al. 2022). Haruna et al. (2020) further reported that cover crops reduced bulk density by ~4.00%, increased macroporosity by ~33.0% and enhanced infiltration by up to 6.29%, resulting in nearly 96.0% lower soil loss than bare soil. Intercropping systems involving soybean, chickpea, maize, sorghum, pigeon pea, cowpea and groundnut improve soil moisture conservation through continuous ground cover, complementary rooting patterns and reduced evaporation losses (Mandal et al. 2013; Singh et al. 2020; Reddy et al. 2025).

In arid and semi-arid regions, limited rainfall and high evaporation rates (1500–1900 $mm\,year^{-1}$) often lead to salt accumulation and sodic conditions in Vertisols, resulting in severe structural degradation, low microporosity and very low saturated hydraulic conductivity (< 1.0 $cm\,h^{-1}$). Under such conditions, gypsum application is widely recognized as an effective amendment for improving the physical condition and productivity of Vertisols (Bello et al. 2021). Gypsum supplies soluble calcium ($Ca^{2+}$), which replaces exchangeable sodium ($Na^{+}$) and dispersive cations on the soil exchange complex, thereby promoting clay flocculation, improving aggregation and reducing dispersion (Hulugalle, McCorkell, et al. 2010; Rashmi et al. 2018; Niaz et al. 2023). Consequently, porosity, infiltration and water-holding capacity increase, while runoff and erosion losses decrease (Rahina et al. 2020; Dong et al. 2022; Rashmi et al. 2025). Research further indicates that gypsum-amended Vertisols retain relatively higher soil moisture during dry spells, resulting in more uniform water availability and reduced risks of surface sealing and rapid desiccation (Rashmi et al. 2024). Improved aggregate stability also reduces the intensity and width of shrinkage cracks formed during drying cycles (Painuli et al. 2017). Similarly, Somasundaram et al. (2016) reported that gypsum application produced the smallest crack volume in Vertisols. Long-term gypsum application improves root penetration, nutrient uptake and crop productivity while enhancing resilience under rainfed and moisture-stressed conditions (Shah et al. 2017). Overall, gypsum effectively ameliorates sodicity and improves moisture conservation, structural stability and crack resistance in Vertisols.

Nutrient management influences shrinkage and cracking in Vertisols mainly through its effects on crop growth, root biomass, aggregation, soil moisture dynamics and cation balance rather than through direct modification of clay mineralogy. Balanced fertilization improves canopy cover and root proliferation, thereby reducing surface evaporation and moderating drying intensity, which lowers shrinkage stress and crack development (Hati, Swarup, et al. 2007; Bandyopadhyay et al. 2010). Integrated application of organic and inorganic fertilizers further enhances aggregation, pore continuity, biological activity and structural resilience during wetting-drying cycles. Organic amendments such as farmyard manure (FYM), compost, green manure and vermicompost increase soil organic matter, aggregation, porosity and infiltration, thereby improving soil moisture conservation and reducing crack intensity during dry periods (Murphy 2015). Balanced application of chemical fertilizers also stimulates root growth and canopy development, which help reduce direct evaporation losses and improve aggregate stability (Zhao et al. 2020). However, greater root biomass may increase soil water extraction under prolonged dry conditions. Long-term integrated nutrient management in Central Indian Vertisols



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increased soil organic carbon by 63.6%, improved aggregation and microporosity, and reduced bulk density after 31 crop cycles (Hati et al. 2006). Similarly, Bandyopadhyay et al. (2003) reported reductions in crack depth and crack volume by 9.60% and 54.3%, respectively, under integrated nutrient management. Al-Nasser and Al-Khalid (2020) reported that 6.0% sheep residue reduced the crack intensity factor to 2.25% compared with 3.78% in the control. Likewise, combined application of wheat straw, sawdust, FYM, fly ash and gypsum reduced crack volume by 18.0%–49.0% in Vertisols (Somasundaram et al. 2016), whereas FYM reduced crack width and depth by 30.8% and 21.9%, respectively (Painuli et al. 2017). Overall, integrated nutrient management improves soil-water relationships, reduces runoff and erosion, and enhances the long-term sustainability of Vertisols (Wani et al. 2003).

Closely associated with nutrient dynamics, soil microorganisms play a crucial role in improving structural stability and reducing cracking susceptibility in Vertisols through their effects on aggregation, organic matter transformation, pore continuity and biological binding processes (Rahman et al. 2017). Among microbial groups, arbuscular mycorrhizal fungi (AMF/VAM) are particularly important because their extensive hyphal networks physically bind soil particles and aggregates, thereby improving aggregate stability and reducing slaking and dispersion during wetting–drying cycles (Miller and Jastrow 2000). Bearden and Petersen (2000) reported that AMF inoculation significantly improved aggregate stability, soil water characteristics, macropore stability and drainage in Vertisols, thereby enhancing resistance to shrinkage-related cracking. In addition to physical binding, AMF produce glomalin-related soil proteins (GRSP), which act as biological cementing agents that stabilize macroaggregates, improve pore continuity, and increase resistance to erosion and structural degradation (Neergaard Bearden 2001). Enhanced aggregate stability reduces shrink–swell stresses and minimizes the formation of deep interconnected cracks. Microbial decomposition products, extracellular polysaccharides, fungal exudates and root-fungal interactions further strengthen aggregate cohesion, improve porosity, and enhance infiltration, aeration and moisture retention in Vertisols (Hati, Biswas, et al. 2007; DeCarlo and Caylor 2019).

In addition to agronomical and biological approaches, texture modification involves the deliberate alteration of soil particle-size composition to improve soil physical behaviour, primarily through the incorporation of coarser materials such as sand into clay-rich Vertisols (Mindari et al. 2023; Qusai et al. 2025). Sand incorporation reduces the dominance of active clay minerals responsible for swelling and shrinkage, thereby lowering volumetric changes during wetting–drying cycles. In addition, sand particles create larger and more stable pores, improving hydraulic conductivity, infiltration and aggregate stability while reducing dispersion and surface sealing. By acting as a skeletal framework that restrains clay movement, sand also limits crack width, depth and continuity. However, excessive sand addition may reduce water-holding capacity and nutrient reserves, making optimization of sand proportion critical. Several studies have demonstrated the effectiveness of controlled sand incorporation in reducing shrink-swell behaviour and crack propagation in Vertisols. Alnmr and Ray (2024) reported that incorporation of 10.0%–50.0% fine marine sand significantly reduced shrinkage, liquid limits and plasticity indices, recommending at least 30.0% sand for improved permeability and soil quality. Similarly, Sabari Ramesh and Thyagaraj (2020) observed marked reductions in vertical, lateral and volumetric deformation during repeated wet-dry cycles with 50.0% sand addition. Dulur et al. (2015) further reported that 20.0% sand incorporation improved aggregate stability and reduced soil instability in Vertisols. However, this approach is generally site-specific and has limited large-scale applicability because of practical, economic and logistical constraints associated with modifying soil texture under field conditions.

Agroforestry systems further improve the structural stability of Vertisols through deep and perennial root networks that physically reinforce the soil matrix, create biopores and interrupt crack continuity (Fahad et al. 2022). Continuous root turnover, root exudates and litter inputs enhance aggregation, organic binding and soil strength, thereby modifying shrinkage stress distribution and reducing crack depth and connectivity during drying (Mohanty et al. 2012) (Figure 6). In addition, decomposition of leaf litter, roots and organic residues contributes to humus formation and long-term soil organic carbon accumulation, improving soil structural resilience and moisture retention (Pan et al. 2025). Agroforestry practices also improve infiltration, reduce erosion and enhance water conservation in Vertisols. Several studies demonstrate the effectiveness of tree-based systems in mitigating cracking and improving soil hydraulic functioning. Bursera graveolens- and *Lysiloma divaricatum*-based agroforestry systems increased soil moisture retention, aquifer recharge and reduced soil cracking in semiarid regions (Pulido-Esquivel et al. 2025). Similarly, *Acacia senegal*-based agroforestry improved soil physical and chemical properties in Sudanese Vertisols (Fadl 2012). Shabtai et al. (2014) further reported that conversion of native *Acacia seyal* and *Balanites aegyptiaca* savannah to cultivated land increased swelling, structural instability and cracking in sodic Vertisols. Several agroforestry systems involving species such as neem, babul, khejri, Pongamia, jamun, subabul, tamarind and Prosopis spp. are well adapted to Vertisols and contribute to improving soil stability and resilience under semi-arid conditions (Ramachandra 2010; Ramesha et al. 2025).

Similarly, agri-horticultural systems improve the structural stability of Vertisols through persistent root occupancy, continuous organic residue addition and enhanced aggregation. Long-term inputs of leaf litter, prunings and root biomass increase soil organic matter, improve aggregate cohesion and elasticity and reduce shrinkage stress concentration, thereby moderating crack width and continuity. Deep-rooted perennial fruit trees such as mango (*Mangifera indica*), custard apple (*Annona squamosa*), ber (*Ziziphus mauritiana*), aonla (*Emblica officinalis*) and tamarind (*Tamarindus indica*) improve nutrient cycling, subsoil aggregation, infiltration and moisture retention while reducing bulk density and swelling behaviour. Tree roots also function as natural biodrains during high-rainfall periods, helping maintain a stable subsurface moisture regime and reducing surface cracking through continuous cover and soil organic matter inputs. Intercropping horticultural plantations with legumes such as cowpea, red gram, bengal gram and horse gram further improves nitrogen availability and aggregate stability through enhanced microbial activity in the rhizosphere (Wang et al. 2024).



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**FIGURE 6** | A conceptual representation of the role of the agroforestry system to reduce soil cracking in Vertisols.

In situ residue recycling beneath tree canopies additionally minimizes evaporation losses and moderates soil temperature fluctuations. Meena et al. (2023) reported that the horti-pastoral system (*Emblica officinalis* + *Cenchrus ciliaris*) reduced runoff and soil loss by 7.50% and 13.9%, respectively, in ravine Vertisols. However, agri-horticultural systems may initially increase runoff and erosion during the establishment phase, indicating the need for complementary soil and water conservation measures. Overall, agri-horticultural systems provide a sustainable nature-based approach for improving soil stability, reducing cracking and conserving moisture in Vertisols.

**(ii) Hydrological and land-configuration practices:** Hydrological and land-configuration practices play a crucial role in regulating soil moisture dynamics, minimizing runoff and erosion, and reducing shrinkage-induced cracking in Vertisols. Because Vertisols are characterized by low infiltration rates, poor drainage and pronounced shrink-swell behaviour, the adoption of suitable land-configuration measures is essential for maintaining favourable soil-water relations and sustaining crop productivity under rainfed conditions. These practices primarily function by improving infiltration, enhancing temporary water storage, and reducing rapid wetting-drying cycles that intensify crack formation (Table 3).

Among hydrological and land-configuration practices, the Broad-Bed and Furrow (BBF) system is one of the most effective approaches for managing Vertisols with 1%–2% slopes that are susceptible to seasonal waterlogging and severe cracking during dry periods. The system consists of broad raised beds (1.0–1.2 m wide and ~15 cm high) separated by narrow furrows, which improve surface drainage, infiltration and moisture redistribution while maintaining favourable root-zone aeration (Hulugalle, Weaver, and Finlay 2010; Erkossa et al. 2014) (Figure 7b). However, the effectiveness of BBF largely depends on rainfall and soil depth. It performs best on deep Vertisols in medium- to high-rainfall regions (≥ 750 mm), where efficient removal of excess water enhances crop establishment and productivity. In contrast, its benefits are limited in shallow or low-rainfall (< 500 mm) Vertisols because furrows may accelerate soil moisture loss rather than conserve it (Pathak et al. 2011). Field studies demonstrated that BBF systems significantly reduced runoff, soil loss and peak runoff rates compared with flat planting. For example, permanent raised beds with residue retention in Ethiopian Calcic Vertisols reduced runoff and soil loss by 53.0% and 79.0%, respectively (Araya et al. 2011). According to Garg et al. (2022) results from long-term field experiments revealed that compared with flatbed cultivation, raised beds reduced surface runoff by 35–40 mm in Vertisols and decreased soil loss by



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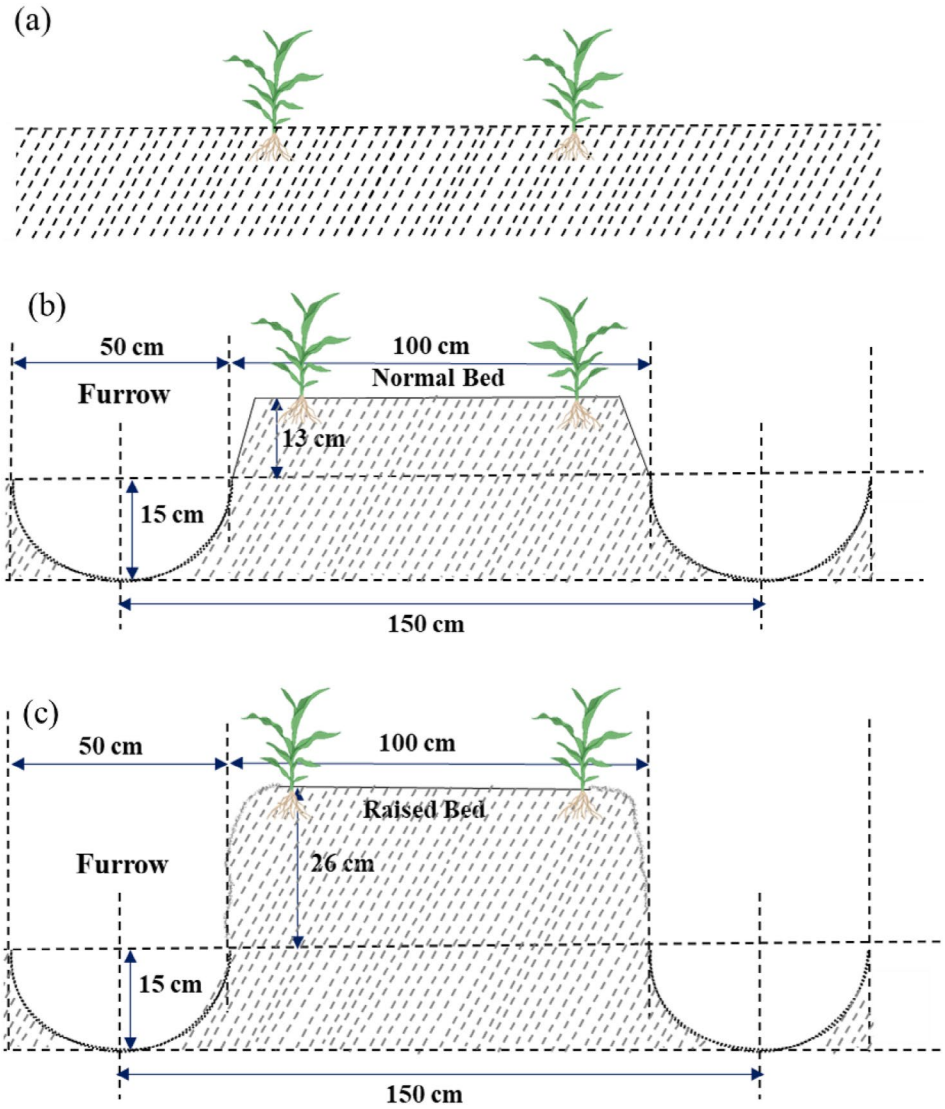



**FIGURE 7** | Showing (a) flat on grade (b) normal broad bed and furrow system (BBF), and (c) raised BBFs for sorghum cultivation in Vertisols.

30.0%–60.0% in semi-arid tropics (Figure 7a,c). In high-rainfall, waterlogged highlands (Ethiopia), drained raised-bed cropping on BBF and ditch drainage gave about 33.0% and 22.0% more grain yield than flat treatment, respectively, and durum wheat grown on BBF with recommended fertilizer gave a 65.4% yield advantage over ridge-and-furrow but further improvements in surface drainage are needed to effectively reduce seasonal waterlogging in Vertisols (Welderufael and Woyessa 2009). Furthermore, BBF systems improved chickpea yield and water-use efficiency by 17.3% and 29.1%, respectively (Kumawat et al. 2025), whereas soybean yield and rainwater-use efficiency increased by 27.1% and 26.3% (Asewar et al. 2017). By enhancing infiltration and maintaining more balanced moisture conditions, BBF systems reduce desiccation stress and restrict deep crack development in Vertisols (Anantha et al. 2021).

Closely related to BBF systems, ridge and furrow (RF) configurations also improve water distribution and soil moisture conservation particularly deep Vertisols under semi-arid to sub-humid climates, where seasonal rainfall is sufficient to create temporary waterlogging but not prolonged flooding. (Figure 8). In RF systems, alternating ridges and furrows facilitate rainwater infiltration through furrows while maintaining relatively aerated ridges for seed placement and root growth (Rao et al. 2022). However, their effectiveness depends strongly on rainfall conditions. In high-rainfall regions, ridge planting minimizes waterlogging, whereas in low-rainfall environments (<800 mm), furrow or flat planting better conserves soil moisture (Pathak et al. 2011). Compared with BBF systems, RF systems reduced runoff and soil loss by 20.2% and 24.1%, respectively (Erkossa et al. 2005). Moreover, integration of RF systems with residue mulching or intercropping further improves resilience against both drought and excess moisture conditions (Kadry and El-Gawad 2023). Ridge-furrow planting combined with biodegradable film mulching increased soil water storage by 26.3 mm, yield by 2665 kg ha$^{-1}$, and water-use efficiency by 51.6% compared with flat planting (Ren et al. 2017), whereas Zhang et al. (2024) reported 26.7% higher yield and 25.8% greater water-use efficiency under ridge-furrow cultivation.

In addition to these field-scale configurations, bunding is a mechanical/engineering soil-and-water conservation measure widely used in rainfed and semi-arid tracts, including Vertisols, where the water that would otherwise run off the field is held



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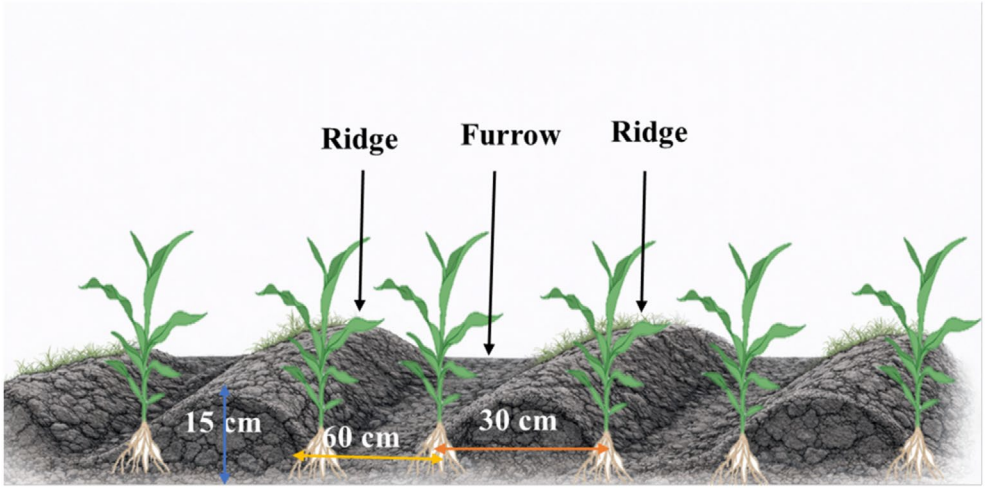



**FIGURE 8** | Representing ridges and furrows system of crop cultivation in arid and semi-arid tropics regions.

back long enough to percolate into the soil profile, where crops can use it. Its importance is especially high in dryland farming because in erratic-rainfall regions without irrigation, losing rainwater to runoff along with the fertile topsoil it carries is one of the most persistent threats to agricultural productivity. The bunding has several benefits that include: bunds create micro-catchments that intercept runoff, increase rainfall capture, promote water storage and deep infiltration, and reduce gully formation, supporting more reliable crop growth even in low-rainfall years. Contour bunding, commonly practiced on 2.0%–6.0% slopes under low and erratic rainfall conditions, slows the movement of runoff water, thereby increasing infiltration and maintaining a more uniform soil moisture profile (Rao et al. 2022, 2024). Such moderated soil moisture conditions help reduce shrinkage intensity and crack formation. When contour bunding is combined with ridge-furrow systems, improvements in soil structure, nutrient retention, root development, and water-use efficiency become more pronounced (Birhanu et al. 2022). Supporting this, Singh et al. (2025) reported reductions of 44.0%–52.0% in soil loss and 36.0%–46.0% in runoff through contour bunding and terracing. Similarly, graded bunding is recommended for Vertisols with steeper slopes (6.0%–10.0%) and rainfall exceeding 600 mm, especially where infiltration rates are $< 8\,mm\,h^{-1}$ (Pathak et al. 2011). Unlike contour bunds, graded bunds include a gentle gradient (0.2%–0.4%) that safely drains excess runoff while preventing prolonged water stagnation. Rao et al. (1998) observed reductions of 74.2% in runoff and 85.0% in soil loss in Karnataka Vertisols, whereas Rao et al. (2022) reported substantial yield improvements in pigeon pea, horsegram, black gram, green gram and chickpea under graded bunding systems.

Complementing these runoff-control measures, several in situ rainwater-harvesting practices further improve soil moisture conservation in Vertisols. Scoop or pitting techniques involve constructing small crescent-shaped basins across slopes to intercept and retain runoff water (Kimaru-Muchai et al. 2021). These micro catchments maintain more stable soil moisture conditions, thereby reducing severe wetting–drying cycles and associated crack formation. In Karnataka Vertisols, scoops reduced runoff and soil loss by 50.0% and 65.0%, respectively, while increasing sorghum yield by 12.0% (Pathak et al. 2011). Likewise, compartmental bunding divides fields into small compartments using low earthen bunds that temporarily retain runoff and increase infiltration (Rao et al. 2022). Improved root growth and moisture retention further increased sorghum yield by 22.8%–25.6% compared with flat beds (Patil 2016). Similar benefits are achieved through dead furrows and tied ridges, which function as in situ water-harvesting structures within the field. Dead furrows intercept rainfall and promote gradual infiltration, whereas tied ridges create interconnected basins that store water for longer durations. Consequently, these systems improve soil moisture distribution, reduce runoff and nutrient losses, and moderate shrinkage-induced cracking.

Beyond field-scale practices, landscape-level hydrological interventions such as subsurface dams, sand storage dams, farm ponds, and check dams play a supportive role in maintaining stable moisture regimes in Vertisol regions. Subsurface dams conserve water below the soil surface by installing impermeable barriers across valleys or aquifers (Onder and Yilmaz 2005) (Figure 9a). These structures improve groundwater recharge and reduce deep drying fronts, thereby minimizing the development of large persistent cracks (Dortaj et al. 2020; Shen et al. 2025). In the Swarnamukhi River basin, subsurface dams increased groundwater levels by 1.44–1.80 m and substantially reduced seasonal groundwater fluctuations (Raju et al. 2006). Frankl et al. (2011) similarly demonstrated reduced gully erosion and improved groundwater recharge in Ethiopian Vertisols following installation of subsurface geomembrane dams.

Sand storage dams contribute to groundwater conservation by trapping runoff and sand in ephemeral stream channels, creating subsurface reservoirs that store water within the accumulated sand (Hanson and Nilsson 1986) (Figure 9b). Although these structures are not directly constructed within Vertisol fields, they improve groundwater recharge and supplemental water availability in Vertisol-dominated landscapes (Aerts et al. 2007; Neufeld et al. 2021). Lazurko et al. (2024) reported that sand dams increased annual water availability from 6.50 to 10.9 months in semi-arid Zimbabwe, thereby strengthening drought resilience and indirectly reducing excessive drying and seasonal cracking in nearby Vertisols. Farm ponds and check dams further complement these hydrological interventions by



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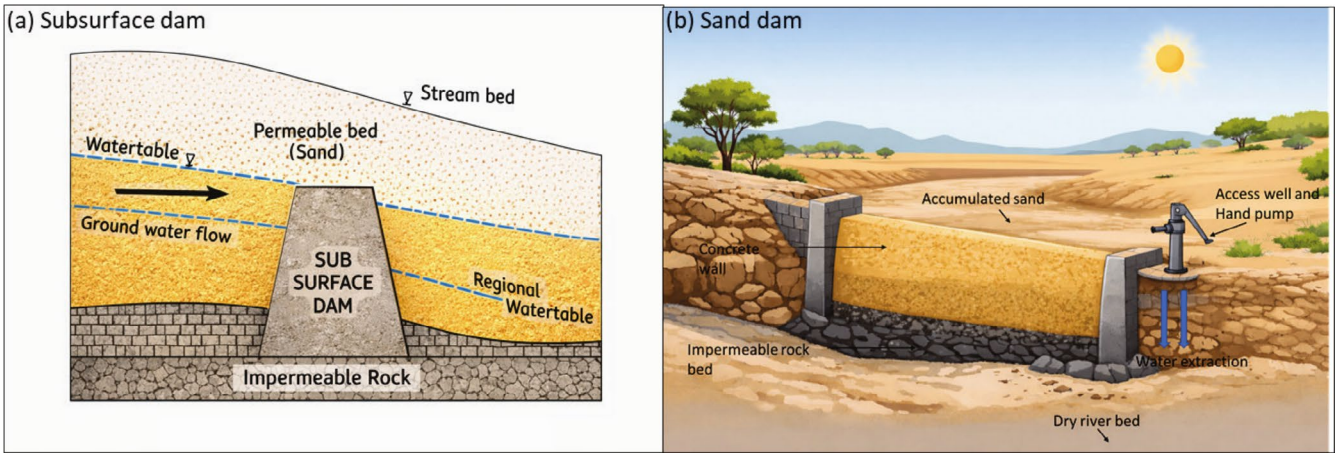



**FIGURE 9** | Conceptual diagram of (a) subsurface and (b) sand dams for groundwater storage in seasonal streambeds.

providing supplemental irrigation during critical crop growth stages. Farm ponds harvest surplus runoff and reduce repeated wetting-drying cycles that intensify cracking (Baruah and Hazarika 2025). Stored runoff from ponds and community tanks improves groundwater recharge and supports life-saving irrigation (Garg et al. 2020; Anantha et al. 2021). Similarly, check dams reduce runoff, erosion and land degradation while increasing groundwater recharge and crop productivity (Polyakov et al. 2014; Piton et al. 2017). Additionally, Garg et al. (2012) reported that such water-management interventions improved water availability by 10.0%–30.0% in semi-arid regions.

Finally, conservation trenching represents another effective runoff-management practice for degraded lands and sloping areas. Continuous contour trenches, staggered contour trenches, and composite trenches intercept runoff, improve infiltration and enhance soil moisture storage (Kurothe et al. 2014; Ali et al. 2024). Among these, staggered contour trenches are particularly effective for degraded lands because they improve vegetation growth, reduce erosion and enhance ecosystem productivity. Alem et al. (2020) reported improved soil moisture and biomass production under trench systems, whereas Shaikh et al. (2021) and Beckers et al. (2013) highlighted their role in improving residual moisture availability and protecting crops against runoff and soil erosion in arid and semi-arid environments. Overall, hydrological and land-configuration practices complement one another by improving infiltration, conserving soil moisture, reducing runoff and erosion, and moderating shrinkage-induced cracking. Through integrated regulation of soil-water dynamics, these practices substantially enhance the structural stability, productivity and long-term sustainability of Vertisols under rainfed conditions.

## 6 | Conclusion

Vertisols are inherently prone to severe surface cracking due to their smectite-rich clay mineralogy and pronounced shrink-swell behaviour under fluctuating moisture regimes. The present review clearly demonstrates that soil and water management practices play significant roles in mitigating surface cracking and improving the overall hydrological functioning of Vertisols. Across diverse agro-climatic regions, these management interventions substantially reduced crack depth, width, length, area and volume, thereby limiting crack continuity and seasonal persistence. Among the practices, organic amendments, residue retention, conservation agriculture, integrated nutrient management, gypsum application and conservation land configurations consistently emerged as the most effective strategies for improving soil aggregation, stabilizing soil structure, moderating shrink-swell stresses and enhancing moisture retention. These improvements not only reduced cracking intensity but also contributed to lower runoff and soil loss, increased infiltration and enhanced water-use efficiency. In contrast, poorly managed tillage systems, inadequate residue retention and puddling practices often intensified cracking and structural degradation, highlighting the importance of appropriate soil and water management under Vertisol conditions. The review further indicates that the effectiveness of management practices is strongly influenced by climatic conditions, residue availability, cropping systems and soil structural characteristics. Therefore, adopting integrated, location-specific and climate-responsive management strategies is essential for sustaining soil health, improving crop productivity, and enhancing the resilience of Vertisols in semiarid and arid regions.

Future research should focus on long-term field evaluations across diverse agro-ecological environments to better understand the temporal dynamics of crack formation and the cumulative impacts of management practices on soil hydrology and ecosystem functioning. Greater emphasis is also needed on mechanistic studies linking soil cracking with infiltration, preferential flow, nutrient transport and carbon dynamics under changing climate scenarios. In addition, the integration of advanced tools such as remote sensing, digital soil mapping, and process-based modelling may improve the monitoring, prediction, and site-specific management of cracking behaviour in Vertisols. Collectively, these efforts will support the development of sustainable and resilient soil and water management strategies for Vertisol-dominated agroecosystems.

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The authors used an AI-assisted design tool (Canva AI) to enhance the visual presentation of Figures 3–5. The original figures, including their scientific concepts, structure, content, and interpretation, were developed entirely by the authors using Microsoft PowerPoint. Canva AI was subsequently used only to improve graphical appearance, readability, and visual clarity. No AI tools were used for data generation, data analysis, interpretation of results, literature review, or preparation of the scientific content of the manuscript. The authors carefully reviewed and verified all figure elements and take full responsibility for the accuracy and integrity of the manuscript.

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The authors have nothing to report.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

No datasets were generated or analysed during the current study.

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